

Foreign Giants’ Broad Shoulders: The Causal Effect of Foreign Ideas on U.S. Innovation*

Tarek Hassan,[†] Aakash Kalyani[‡] and Juan Cruz Lopez Del Valle[§]

October 1, 2026

Abstract

We study the role of foreign ideas in U.S. innovation by constructing a directed “genealogy” of technologies from the full text of 13 million patents filed between 1894 and 2024. By tracing which earlier technologies appear in the founding patents of new ones, we recover a large-scale input–output network of technological recombination that identifies the foreign technologies that serve as inputs into innovation in the United States. We report four main findings. First, foreign ideas are pervasive: 56% of U.S. technologies build on at least one foreign input, and the most fertile U.S. technologies rely on foreign inputs most heavily. Second, foreign ideas have large causal effects on U.S. innovation. Using the historical settlement of migrants as a source of exogenous variation in local ancestry, we show that U.S. counties more exposed to knowledge from a given foreign country absorb substantially more technologies from that country and, in turn, invent significantly more U.S. technologies building on them. Third, foreign ideas shape not only the level but also the direction of technical change: U.S. counties are far more likely to invent the specific technologies whose foreign inputs are already in their local idea stock, and the changing origins of foreign ideas accompany the shift of American innovation from its historical heartland toward the coasts and the Sunbelt. Fourth, these effects leave a lasting imprint on local labor markets: decades later, counties that absorbed more of an origin’s ideas host more jobs working with those ideas, and more still working with the U.S. technologies descended from them.

*We thank Daron Acemoglu, Nick Bloom, Marcela Carvalho, Kevin Lang, Josh Lerner, Ezra Oberfield, Pascual Restrepo, Ahmed Tahoun, and Chris Tonetti, as well as seminar participants at Harvard Business School and the Federal Reserve Bank of St. Louis, for helpful comments. The views expressed here are solely those of the authors and do not necessarily represent those of the Federal Reserve Bank of St. Louis or the Federal Reserve System.

[†]**Boston University**, 270 Bay State Rd, Boston, MA 02215; thassan@bu.edu.

[‡]**Federal Reserve Bank of St. Louis**, One Federal Reserve Bank Plaza, St. Louis, MO 63102; aakash.kalyani@stls.frb.org.

[§]**Boston University**, 270 Bay State Rd, Boston, MA 02215; jczlv@bu.edu.

The notion that ideas build on ideas is perhaps the oldest and most durable proposition in the study of economic growth, running from [Schumpeter \(1911\)](#) through [Romer \(1990\)](#) and [Aghion and Howitt \(1992\)](#). But if new knowledge is made by recombining existing knowledge, and existing knowledge diffuses slowly across space ([Griliches, 1957](#); [Jaffe et al., 1993](#); [Keller, 2002](#)), then where an inventor works shapes what she is able to invent. Together, the two propositions imply that innovation should be understood as production: new ideas are produced from old ones, and their manufacture depends on which input ideas are locally at hand.

Although both propositions are widely accepted, their implications have proven difficult to study directly, because the objects they involve are usually assumed to be unobservable. In Krugman’s ([1991](#)) famous formulation, knowledge flows “are invisible; they leave no paper trail by which they may be measured and tracked.” As a result, we know little about the production structure of ideas—which technologies are inputs into which new technologies—or about its geography—which of those ideas are available in a given place at a given point in time.

In this paper, we develop a method that reads both objects directly from the full text of patents. Applying it to thirteen million USPTO patents filed between 1894 and 2024, we construct a directed genealogy of technologies—an input–output network recording which existing technologies served as inputs into the creation of each new one—and, from the location of the inventors who mention each technology, the stock of ideas available in each U.S. county at each point in time. We then use these two objects to quantify how local access to foreign ideas shapes American innovation and the local job opportunities it creates.

We document four main findings. First, the majority of technologies developed in the United States build on at least one foreign input technology—and the innovations with the most children rely on foreign ideas most heavily. Second, this dependence is causal: U.S. counties with greater exposure to a foreign country’s knowledge—instrumented with the historical settlement of its migrants—absorb more of its technologies. A 1% increase in absorption of these ideas from a given foreign country raises local innovation building on those foreign technologies by 1.4%. Third, local idea stocks shape not just how much a place invents but *what* it invents: U.S. counties are far more likely to develop precisely those new technologies whose inputs are already locally

absorbed. Fourth, these effects carry over to local labor markets: a 1% increase in the absorption of ideas from a given origin country raises the number of local job postings that mention the U.S. technologies descending from those ideas by 3.5%.

We begin by constructing a panel of technologies and their countries of origin. The key step, which we take from [Kalyani et al. \(2025\)](#), exploits the fact that when a new technology comes into the world, so does the language used to refer to it. We start from the universe of technologies described in Wikipedia, associate each with the one- and two-word phrases that name it, and search for these phrases in the full text of patents. Whereas most existing data on innovation describe patents—their counts, their citations, their textual similarity—this procedure makes the *technology itself* the unit of analysis: for example, we can follow polyvinyl chloride (PVC) or Bluetooth as an object in its own right, observing each patent that works with it, wherever and whenever it is filed. Applying this procedure to our patent sample, we identify 19,542 technologies, and for each record the first country in which patenting surpasses a minimal threshold—its country of origin—and the year in which it does so—its birth year. Of these, 1,386 (roughly 7%) first emerged abroad, with Japan, Germany, and the United Kingdom accounting for the majority, spanning the century’s defining innovations from the electron microscope (Germany, 1939) to cloud computing (Sweden, 2008). Because countries differ in their propensity to patent in the United States, we do not interpret these counts as measures of national inventiveness; most of our empirical work compares outcomes within technology or within origin country, absorbing such differences.

We then construct the genealogy. Because our data identify technologies rather than individual documents, we can observe which technologies appear together in the same patents—and, in particular, which existing technologies appear in the early patents that found a new technology. For each technology, we define its founding patents as those filed up to five years after its birth. We then say that an older technology is a *parent* of a newer one if it is mentioned in at least ten percent of the newer technology’s founding patents. Each such link records an input into the production of a new idea. For example, when the earliest patents developing cling film, the floppy disk, and high-density polyethylene all work with polyvinyl chloride (PVC), we conclude that PVC was an ingredient in their creation. Applying this definition to all 19,542 technologies yields a directed

input–output network of U.S. technological development since 1894—to our knowledge, the first of its kind.

The genealogy reveals that U.S. innovation draws pervasively on foreign ideas. The average U.S. technology has 48 parents, of which 3 first emerged abroad, and 56% have at least one foreign parent. Technology fertility is highly skewed: half of all technologies have no children within ten years of their birth and the median among the rest has 3, while the most prolific, such as PVC and Bluetooth, spawn hundreds. Strikingly, it is precisely these U.S. technologies with the most children—television, the floppy disk, big data—that rely most heavily on foreign parents. The ideas that transform American technology are thus disproportionately built on the shoulders of “foreign giants.”

We then measure the local stock of ideas. We say a technology is *absorbed* in a U.S. county once it is mentioned in patents filed by local inventors—direct evidence that local inventors have begun working with it, and thus that the technology has entered the U.S. county’s stock of knowledge. Aggregating across technologies from a given origin country yields a U.S. county-by-origin country panel of the local absorption of domestic and foreign ideas.

Having measured each county’s stock of ideas, we ask a simple question: are some U.S. locations better at absorbing foreign ideas than others? We first document that the local absorption of foreign ideas is strongly predicted by the U.S. county’s ancestry from the technology’s country of origin. To establish that this link is causal, we instrument for ancestry using the historical push–pull design of [Burchardi et al. \(2019\)](#), which exploits quasi-random variation in where successive waves of migrants settled. Relating the technologies a U.S. county absorbed between 1980 and 2019 to its ancestry in the 1980 census, we find that a 1 percent increase in a county’s ancestry from a given origin raises its absorption of that origin’s technologies by about 0.46 percent. Counties with a disproportionate population of German ancestry, for example, import far more ideas from Germany and work with them more frequently in local patenting. The ethnic composition of a county, inherited from a century of migration, thus endows it with a comparative advantage in absorbing knowledge from specific origins.

Combining this result with the genealogy of technologies then allows us to test the proposition

with which we began—that new ideas are made from old ones, and that their manufacture depends on which input ideas are locally at hand. Our genealogy makes this test possible because, for every foreign technology, we observe both where it was absorbed and where and when its descendants were born. Specifically, we relate the number of new technologies a county develops after 1980 that build on parents from a given foreign country to the U.S. county’s absorption of that foreign country’s ideas before 1980, again using the settlement history of migrants for identification. A mechanical concern is that working on a parent and inventing its child are difficult to distinguish in patent text; we therefore measure what we call *non-generative* absorption, discarding any patent that itself mentions a child technology. Non-generative absorption captures local inventors who learned to work with a foreign technology—figuring out, say, how to color PVC—without founding any new technology of their own.

It is precisely this “non-generative” mastery that predicts the future: a 1% increase in a county’s non-generative absorption of technologies from a given origin raises its subsequent invention of their children by 1.4%. One might have thought that ideas travel through kinship ties on demand—that a county with many German descendants can draw on German ideas whenever it needs them. The balance of the evidence suggests otherwise. Allowing ancestry itself, the foreign-born population, recent immigration, foreign direct investment, or demand for information about the origin country to affect innovation directly leaves the effect of absorption essentially unchanged. It thus appears that ideas must be worked with locally before they can be recombined into something new. This is, to our knowledge, the first direct evidence at scale of inventors standing on the shoulders of giants: the places that absorb an idea today are the places that build on it tomorrow.

The absorption of foreign ideas also leaves a lasting imprint on local labor markets. In the text of 350 million online job postings from 2010–2023, U.S. counties that absorbed more of a foreign country’s technologies before 1980 host more jobs working with those imported technologies today—and even more jobs working with the American technologies that descend from them, with an effect about a quarter larger. Ideas absorbed generations ago thus still shape the job opportunities on offer in a U.S. county today, less through the imported technologies themselves than through the domestic innovations they made possible.

Our final set of results sharpens the analysis from country aggregates to individual technologies: we ask not how much a U.S. county invents, but *which* technologies it invents. For each American-born technology and each U.S. county, we measure whether that particular technology’s parents were already available locally in the preceding five-year period. Counties with a technology’s specific inputs at hand are far more likely to be among its inventors.

The magnitudes are large. Inventing any given technology in any given county is a rare event—the outcome is one in less than one percent of technology–county–period cells. Local availability of a technology’s key foreign inputs raises the chance that the county is among its inventors by about 5 percentage points in the raw data, and by substantially more, 23 percentage points, once we instrument for availability using, again, the settlement history of migrants. The local stock of foreign ideas thus determines not only the volume of American innovation but its direction. Consistent with this, we show that the changing origin countries of foreign ideas after 1980—away from Europe, toward Asia—align with the marked shift of American innovative activity away from the heartland and toward the coasts.

We make three main caveats to our interpretation. First, our estimates identify how the stock of foreign ideas allocates innovation *across* U.S. counties; they do not measure the aggregate gain to the United States from foreign knowledge, since absorption in one location may in part displace innovation that would have occurred elsewhere. Second, our measurement relies on patent text and thus misses technologies that are never mentioned in patents. This does not restrict our data to patentable inventions, however: a technology enters our data whenever patents work with it, whether or not it is itself patentable. As a case in point, “software” is one of the technologies in our sample. Third, we assign each technology to the first origin country in which it generates substantial patenting at the USPTO. We may therefore misattribute technologies from countries with a low propensity to file there—a concern most relevant early in the sample and for countries such as China.

Related literature. A large literature seeks to understand how ideas build on ideas. In growth theory, new knowledge is produced by recombining existing knowledge ([Schumpeter, 1911](#); [Romer, 1990](#); [Aghion and Howitt, 1992](#); [Weitzman, 1998](#); [Jones, 2009, 2023](#)). A growing empirical literature

measures this process with networks constructed from patent data (Jaffe, 1986; Jaffe et al., 1993; Bloom et al., 2013; Ahmadpoor and Jones, 2017). Most recently, Acemoglu et al. (2016) and Liu and Ma (2026) measure citation flows between technology classes, while Kelly et al. (2021) link patents through textual similarity, identifying patents that are linguistically novel relative to the past and influential on the future.¹ In each case, the network’s nodes are categories the patent office itself adjudicates or records—patents, classification codes, named inventors. Our contribution is to make the *technology itself* the node: an object with a specific economic meaning, to whose development many patents, inventors, and firms contribute. Identifying each technology through the specific language coined to describe it lets us equip each node with extensive data: a birth year, a country of origin, the locations and jobs in which it is applied—and, crucially, the parent technologies that appear in its founding patents. The resulting network is, to our knowledge, the first direct empirical counterpart to the production function of technologies that the theoretical literature posits: a directed graph recording which specific technologies entered the creation of which others, at scale, and over more than a century.²

Because each node in this network carries a birthplace and a geography of use, we can go beyond measuring the aggregate production structure of ideas and identify its consequences for local economies. In that sense, we contribute to a large literature that studies how innovation shapes local economies and local labor markets (Van Reenen, 1997; Moretti, 2010; Kline et al., 2019; Kalyani et al., 2025). Relative to this work, we provide causal estimates of the effects of a specific and previously unmeasured input into local innovation—the absorption of foreign ideas—on the invention of new technologies and the job opportunities they create, and show that these effects compound over decades as absorbed technologies spawn domestic children.

We also add to an empirical literature that uses natural experiments to show that access to existing knowledge shapes subsequent invention (Furman and Stern, 2011; Moser and Voena, 2012; Williams, 2013; Galasso and Schankerman, 2015; Sampat and Williams, 2019; Hegde et al., 2023; Gross, 2023; Iaria et al., 2018). Each of these studies traces one change in access, in one

¹See Hall et al. (2001), Young and Kuhn (2016), and Marx and Fuegi (2020) for related data resources.

²Oberfield (2018) and Acemoglu and Azar (2020) model how input–output networks form when producers choose their inputs and how the resulting structure shapes productivity; the genealogy is an observed counterpart in which the inputs are existing technologies and the producers are inventors.

field, through citations to the affected work. We instead observe which technologies are available to inventors in every U.S. county at every date, and trace their effect on the specific technologies subsequently founded on them.

A third literature shows that immigrants raise innovation in receiving regions (Kerr, 2008; Hunt and Gauthier-Loiselle, 2010; Moser et al., 2014; Bernstein et al., 2026; Posch et al., 2026; Terry et al., 2026), and more broadly that co-ethnic networks transmit knowledge and facilitate economic exchange across borders (Burchardi et al., 2019; Prato, 2025). We contribute evidence on a distinct margin: co-ethnicity determines not only how much a place innovates but *what* it innovates, by governing which foreign technological lineages local inventors can build on.

Finally, a large literature studies the direction of technical change. Following Hicks (1932) and Acemoglu (1998, 2002), this literature emphasizes the demand side of innovation: inventors direct their efforts toward technologies with larger markets or scarcer substitutes, whether the market size derives from demographics (Acemoglu and Linn, 2004), factor and energy prices (Popp, 2002; Hanlon, 2015; Aghion et al., 2016), labor scarcity (Acemoglu and Restrepo, 2022), or the conditions of leading markets (Moscona and Sastry, 2025). We contribute evidence on the supply side of the same question: holding the demand for a technology fixed, places are more likely to invent it when its input ideas are locally at hand. We show the direction of innovation is shaped by what its inventors have to work with.

Organization. Section 1 identifies the technologies and their countries of origin and constructs the genealogy that links them. Section 2 measures the local stock of ideas and shows that a county’s ancestry determines which foreign ideas it absorbs. Section 3 estimates the effect of absorbed foreign ideas on local innovation, and Section 4 their effect on local job postings. Section 5 moves to the level of the individual technology and to the changing geography of U.S. innovation. Section 6 concludes.

1 A Genealogy of Technologies

We construct our panel of technologies by minimally adjusting the text-based methodology of Kalyani et al. (2025) and Hassan et al. (2026), with three objectives: to (i) extend the panel further

back in time, (ii) identify the technologies that originate in foreign countries, and (iii) capture technologies that matter as inputs into other technologies.

As in our prior work, we identify all Wikipedia pages that describe a technology by selecting all pages that feature sections whose titles contain words “application(s),” “use(s),” “type(s),” “operation,” “characteristic(s),” “feature(s),” “device(s),” “technical,” or “commercial.” We then pass each of these pages to a large language model (Claude Haiku), asking whether it describes a technological innovation, defined as “a set of specific and applied techniques, products, and processes, as distinct from scientific innovation and from managerial knowledge.” Appendix [A.1](#) reports the full prompt, together with the categories the model assigns and, in Table [A.1](#), examples of pages that passed and failed the filter.

In a second step, we associate each technology with the language used to refer to it. From the title and the first 50 words of the summary of each technology’s page—the part of the page typically used to define and describe the technology—we extract all unigrams and bigrams, giving us a list of candidate phrases for each Wikipedia page. From each list, we discard the phrases that contain stop words, those that appear in the Corpus of Historical American English before 1840 ([Davies, 2010](#)), and those that appear in any patent filed between 1834 and 1894. The first two filters isolate technical language from everyday English; the third removes language already in use in U.S. patents by 1894, the year the United States joined the Paris Convention for the Protection of Industrial Property and foreign inventors began to be consistently identified in U.S. patents. Finally, we make the mapping from phrases to Wikipedia pages unique by entering each remaining phrase into the Wikipedia search engine and keeping only those for which the first technology page returned is the page they came from. Each technology in our data thus consists of the title of a Wikipedia page and a list of unigrams and bigrams uniquely associated with it. In all, the list holds 34,736 technologies and 80,541 phrases.

In a third step, we search for these phrases in the full text of all utility patents filed with the USPTO between 1894 and 2024, counting plural and spelling variants of each phrase. We say a patent “mentions a technology” if its text contains at least two mentions of phrases linked to the technology.

Next, we assign each technology an origin country, $o(\tau)$ —the first country whose inventors file 10 patents mentioning it³—and a birth year, $b(\tau)$ —the first year in which this threshold is reached in any country. Note that this notion of a “birth year” by construction pre-dates the emergence year used in our earlier work, which required 100 mentions and sustained growth thereafter (Kalyani et al., 2025; Hassan et al., 2026). That work asked how new technologies affect the labor market directly, warranting a focus on the point in time when they become commercially impactful. Here, our objective is to also capture technologies that affect the economy indirectly, by serving as inputs into the development of other technologies, regardless of whether they ever develop a large footprint themselves. Locating a technology’s origin also requires its earliest trace, before it has taken off anywhere. The lower threshold is also why our sample of technologies is larger than in our earlier work.

Overall, 19,542 technologies reach this threshold, with birth years between 1894 and 2024; of these, 1,386 (7.1%) were born abroad. Japan (41%), Germany (22%), and the United Kingdom (11%) account for the majority of foreign-origin technologies (Appendix Table A.2).

For the purposes of interpreting these numbers, it is important to keep in mind that foreign inventors are generally under-represented in USPTO data, simply because many foreign inventions are never patented in the United States. In addition, foreign attribution dips during and after both World Wars, when nationals of enemy countries could not file in the United States and the Alien Property Custodian seized enemy-owned patents (e.g.). The wartime restrictions depressed foreign patenting until about 1960 (Gimbel, 1990), and we therefore focus part of our analysis on the post-1960 sample.

Finally, countries differ widely in their propensity to patent in the United States, so we do not interpret these counts as measures of national inventiveness. Most of our estimates compare outcomes within technology or within origin country, and thus absorb such differences.

³When the United States and a foreign country reach this threshold in the same year, we assign the technology to the foreign country.

1.1 Constructing the Genealogy

The advantage of observing technologies directly, rather than patents, their citations, or their similarity, is that we can track which technologies appear together in the same patents. In particular, we can observe which existing technologies inventors mention in the patents that found new technologies.

For any technology τ , define the set of all patents mentioning it,

$$\mathcal{P}(\tau) = \{\text{patents mentioning } \tau\},$$

and for each technology τ' , define its *founding patents* as those filed up to five years after its birth,

$$\mathcal{F}(\tau') = \{\text{patents mentioning } \tau' \text{ filed by } b(\tau') + 5\}.$$

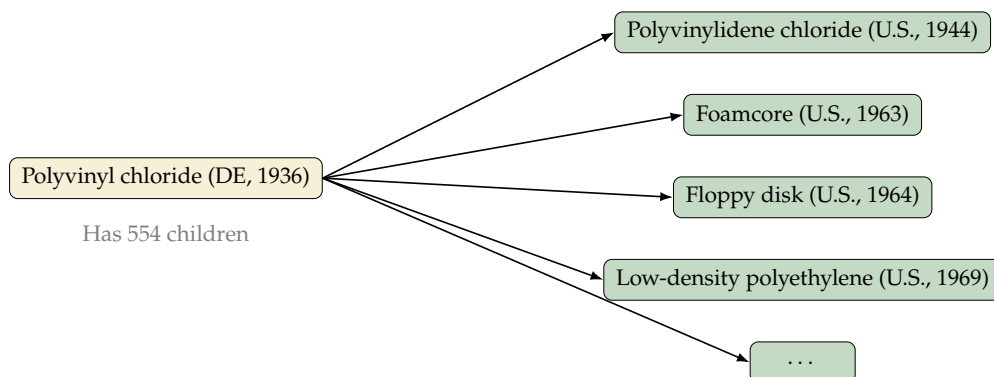
Because the birth year is the year in which patenting first reaches ten in some country, the founding patents include those filed before it. We measure the dependence of τ' on any older technology τ as the share of τ' 's founding patents that also mention τ ,

$$s_{\tau \rightarrow \tau'} = \frac{|\mathcal{F}(\tau') \cap \mathcal{P}(\tau)|}{|\mathcal{F}(\tau')|}, \quad (1)$$

and say that τ is a *parent* of τ' if $s_{\tau \rightarrow \tau'} \geq 0.10$ and $b(\tau) < b(\tau')$; we write $C(\tau) = \{\tau' : s_{\tau \rightarrow \tau'} \geq 0.10 \text{ and } b(\tau) < b(\tau')\}$ for the children of τ . Each such link records an input into the production of a new idea: the inventors founding τ' built on τ . Applying this definition to all pairs of technologies yields a $19,542 \times 19,542$ directed network—a genealogy describing technological development in the United States since 1894.

Figure 1 illustrates the genealogy using polyvinyl chloride (PVC), which first emerged in Germany in 1936. PVC appears in the founding patents of hundreds of later technologies, most of them born in the United States. These include polyvinylidene chloride (1944), the moisture barrier used in food packaging; Foamcore (1963); the floppy disk (1964); and low-density polyethylene (1969), the plastic used in containers and piping. Each of these children is an American technology

Figure 1: A foreign parent technology and its American children.



Notes: The figure shows a foreign parent technology and four of its U.S. children. Equation 1 defines parentage.

built, in part, from a German idea.

1.2 Properties of the Genealogy

The average U.S. technology has 48 parents, 3 of which are foreign, and 56% have at least one foreign parent. Among U.S. technologies born since 1960 the averages are: 59 parents, 3.5 of them foreign, and 63% with at least one foreign parent. Fertility is highly skewed: half of technologies have no children within ten years of their birth, and among those that do, the median has 3, the mean 10, and the 99th percentile 127. Figure 2 plots the distribution.

Table 1 lists, for each birth year, the most fertile foreign technology—the one with the largest number of children—together with its origin country. Reading down the table traces the changing sources of highly impactful foreign ideas over the century. The early decades are dominated by German industrial chemistry: glycolic acid (1911), butadiene (1913), methanol (1925), and sodium dodecyl sulfate (1931) each spawned more than a thousand descendants. Mid-century, the most fertile foreign technologies shift toward European instruments and pharmaceuticals—the electron microscope (1939), cephalosporin (1961), antipsychotics (1966), doxorubicin (1976), tamoxifen (1985). The final decades bring Japanese electronics and Nordic communications: the carbon nanotube (1995), the smartphone (1998), Bluetooth (1999), and cloud computing (2008).

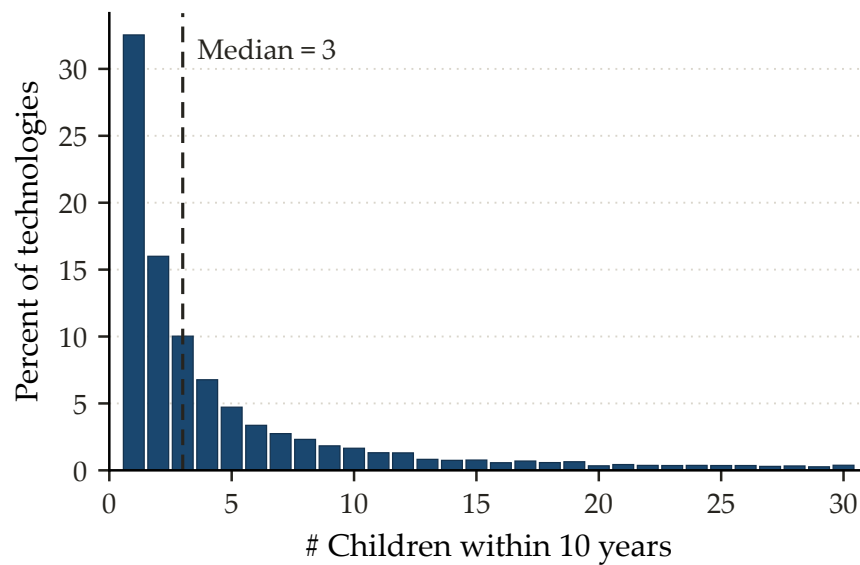
Table 2 reports the American equivalent. The early century is dominated by industrial chemistry, materials, and electrification—formaldehyde (1903), reinforced concrete (1904), the enzyme

Table 1: Most fertile foreign technologies by birth year, 1894–2016.

Year	Technology	Origin	Children	Year	Technology	Origin	Children
1894	Oleum	DE	24	1967	Phenylbutazone	JP	89
1899	Phosphonium	DE	228	1969	Cytostasis	CH	200
1900	Sulfonic acid	DE	44	1970	PBAT	JP	13
1901	Nitrobenzene	DE	229	1971	Octyl glucoside	CH	23
1902	Anthranilic acid	DE	147	1972	Cysteamine	FR	61
1904	Malonic acid	DE	931	1973	Kolliphor EL	GB	195
1906	Tantalum	DE	502	1974	Beta-lactamase	GB	321
1909	Indanthrone blue	DE	19	1975	Fusidic acid	DK	296
1911	Glycolic acid	DE	1,279	1976	Doxorubicin	IT	766
1913	Butadiene	DE	1,009	1977	Thienamycin	GB	20
1917	Carbamate	DE	928	1979	β -Galactosidase	JP	827
1918	Ammonium perchlorate	SE	51	1980	Cefoxitin	JP	164
1920	Hydroponics	DE	104	1981	Sertraline	DE	180
1921	Sodium percarbonate	DE	104	1982	Nifedipine	JP	249
1923	Acetoacetic acid	DE	152	1984	Image file format	JP	426
1924	Carbazole	DE	113	1985	Tamoxifen	GB	440
1925	Methanol	DE	4,317	1986	Cypermethrin	GB	96
1926	Hafnium	NL	197	1987	Ticlopidine	JP	97
1927	Cyclohexanone	DE	334	1988	Ofloxacin	JP	150
1928	Cyclohexane	DE	1,148	1989	Ciprofloxacin	DE	289
1929	DEAE cellulose	DE	164	1990	Glufosinate	DE	145
1930	1,4-Dioxane	DE	1,562	1991	OLAP	JP	50
1931	Sodium dodecyl sulfate	DE	2,602	1992	Fenofibrate	FR	84
1932	Dodecanol	DE	129	1993	Nilutamide	CA	147
1933	Quaternary ammonium	DE	872	1994	SDRAM	JP	303
1934	Hydroxypropyl cellulose	DE	1,272	1995	Carbon nanotube	JP	198
1935	Imidazole	DE	1,240	1996	SIM card	DE	96
1936	Polyvinyl chloride	DE	554	1997	Penciclovir	GB	80
1937	Polystyrene	DE	1,642	1998	Smartphone	FI	848
1938	Triazole	DE	530	1999	Bluetooth	FI	1,220
1939	Electron microscope	DE	295	2000	QR code	JP	136
1940	Optical microscope	DE	164	2001	Graphene	JP	140
1941	Lead(II) nitrate	DE	25	2002	Symbian	FI	73
1943	Metolachlor	GB	55	2003	Etonogestrel	NL	24
1948	Methanesulfonic acid	CH	746	2005	Orbitrap	GB	28
1952	Hydrocyclone	NL	22	2006	Ixabepilone	DE	62
1954	Silver diammine fluoride	CH	11	2007	Pendetide	GB	49
1955	Data store	GB	952	2008	Cloud computing	SE	533
1956	SIMD	GB	610	2009	Pazopanib	DE	77
1957	Peanut oil	GB	61	2010	Argan oil	FR	12
1959	Content-based image retr.	DE	145	2012	MQTT	GB	18
1961	Cephalosporin	GB	315	2013	WebRTC	CN	10
1963	Colchicine	FR	137	2014	Alirocumab	GB	23
1964	Flax	CH	69	2015	Pembrolizumab	GB	59
1965	Enoxolone	GB	56	2016	Ethereum	GB	20
1966	Antipsychotic	CH	289				

Notes: For each birth year, the table reports the foreign-origin technologies with the largest number of children in the genealogy, among technologies with at least ten children. Years in which no such technology was born are omitted. We manually omit *Panicum virgatum* and *Oenothera biennis* which are species of plants and not technologies.

Figure 2: Technology fertility at age 10.



Notes: The figure plots the distribution across technologies of the number of children born within ten years of the technology’s birth. This is among the 9,515 technologies born through 2014 (whose ten-year window is complete in our data) with at least one such child. Bars show the percent of technologies at each count, and the axis is truncated at 30 children (7 percent of technologies lie beyond).

(1907); the interwar decades add plastics, electronics, and the foundations of modern medicine—television (1924), polyethylene (1931), the antibody (1928), the antibiotic (1943). After the war, the information revolution unfolds almost year by year, from data processing (1952) and the computer program (1957) through software (1967), the microprocessor (1972), the database (1975), and the personal computer (1981), interleaved with the rise of molecular biology from the monoclonal antibody (1979) to PCR (1988) and CRISPR (2008). The most recent entries—big data (2010), deep learning (2013), generative adversarial networks (2016)—mark the transition to the data-driven present.

Notably, the list contains many technologies that were not themselves patentable for much of our sample period, including software (1967)—with 8,113 children the most fertile U.S. technology in our data—and communication standards such as the Transmission Control Protocol, Zigbee, and WiMAX. These technologies enter our data because inventors naturally mention them when patenting adjacent, patentable inventions.

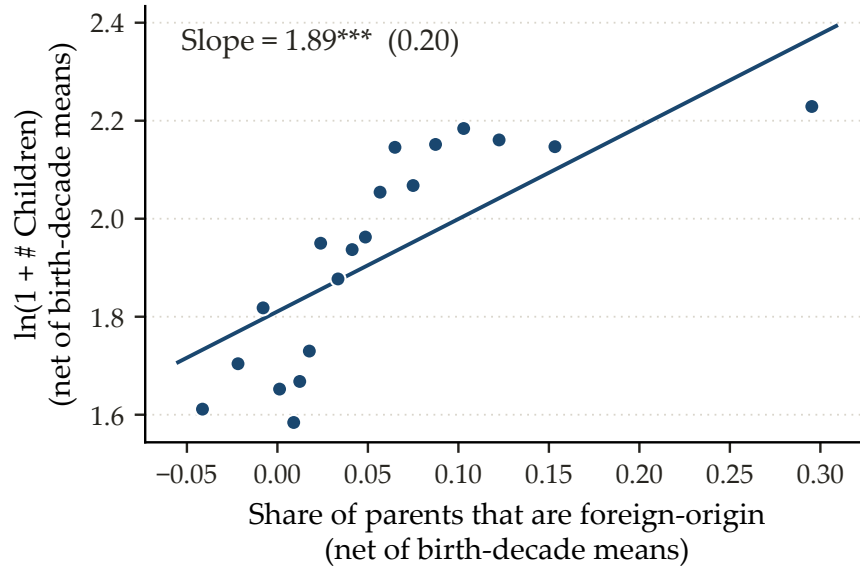
Figure 3 shows a binned scatterplot of the number of children of each U.S. technology against

Table 2: Most fertile U.S. technologies by birth year, 1894–2016.

Year	Technology	Children	Year	Technology	Children
1894	Sulfanilic acid	86	1956	Dimethyl sulfoxide	2,108
1895	Party line (telephony)	64	1957	Computer program	4,342
1897	Rebar	41	1958	Nucleoside	880
1898	Leased line	707	1959	Fetal bovine serum	1,862
1899	Zinc sulfate	168	1960	Data acquisition	855
1900	Iron(II) sulfate	122	1961	Cell culture	1,941
1901	LC circuit	272	1962	Histogram	701
1902	Cellulose acetate	813	1963	Data file	1,355
1903	Formaldehyde	447	1964	Floppy disk	3,906
1904	Reinforced concrete	52	1965	Gel electrophoresis	1,354
1905	Cathodic protection	159	1966	Polynucleotide	1,529
1906	Tricalcium phosphate	246	1967	Software	8,113
1907	Enzyme	3,950	1968	Error message	803
1908	Sub-/superscript	651	1969	Programming language	2,073
1909	Paraformaldehyde	358	1970	Flowchart	3,637
1910	Growth medium	2,015	1971	Tomography	698
1911	Ultraviolet	1,903	1972	Microprocessor	3,854
1912	Glycerol	2,881	1973	Application software	3,366
1913	Argon	1,717	1974	User interface	4,120
1914	Sulfur	528	1975	Database	5,743
1915	Dicalcium phosphate	742	1976	Mutagenesis	1,283
1916	Hexane	2,519	1977	EEPROM	1,666
1917	Ampoule	555	1978	Cloning	1,945
1918	Thermal decomposition	256	1979	Monoclonal antibody	2,226
1919	Ethylene glycol	1,838	1980	Biotechnology	1,177
1920	Isopropyl alcohol	2,088	1981	Personal computer	3,054
1921	Silica gel	1,898	1982	Gene expression	1,536
1922	Protease	2,224	1983	Viral vector	744
1923	Stainless steel	2,026	1984	Streptavidin	890
1924	Television	2,244	1985	Site-dir. mutagenesis	674
1925	Transducer	2,090	1986	Web server	1,394
1926	Capacitor	2,728	1987	Laptop	2,709
1927	Glycine	2,257	1988	PCR	982
1928	Antibody	3,040	1989	Reporter gene	547
1929	Output device	1,742	1990	Flash memory	1,898
1930	Infrared	3,648	1991	Metadata	1,602
1931	Polyethylene	4,583	1992	Embryonic stem cell	321
1932	Intramuscular injection	1,673	1993	PDA	876
1933	Triethylamine	1,665	1994	Workflow	756
1934	Ion exchange	1,390	1995	Middleware	486
1935	Polyester	1,880	1996	Website	1,352
1936	Polyethylene glycol	3,268	1997	Web service	671
1937	Coaxial cable	396	1998	Filgrastim	203
1938	Isopropyl myristate	638	1999	siRNA	340
1939	Epoxy	1,841	2000	Etanercept	198
1940	Sorbic acid	858	2001	Transcriptome	103
1941	Silicone	1,950	2002	Imatinib	247
1942	Telecommunications	1,454	2003	Zigbee	192
1943	Antibiotic	2,973	2004	WiMAX	361
1944	Tetrahydrofuran	1,809	2005	Pegfilgrastim	89
1945	Polylactic acid	782	2006	Sunitinib	142
1946	Streptomycin	1,009	2007	Social media	326
1947	Multiplexer	1,633	2008	CRISPR	112
1948	Growth factor	1,781	2009	Bluetooth Low Energy	89
1949	Chromatography	3,829	2010	Big data	105
1950	Dosage form	1,962	2011	NoSQL	56
1951	Signal processing	1,577	2012	Bitcoin	48
1952	Data processing	3,430	2013	Deep learning	93
1953	EDTA	2,484	2014	Cryptocurrency	45
1954	Memory address	1,474	2015	Talimogene laherp.	21
1955	Tissue culture	1,318	2016	GAN	34

Notes: For each birth year, the table reports the U.S.-origin technologies with the largest number of children in the genealogy, among technologies with at least ten children. Years in which no such technology was born are omitted. A small number of entries are passed over: those whose phrase matches arise from a collision rather than the technology itself, and those naming a phenomenon long predating the year shown.

Figure 3: Foreign parentage and technological influence.



Notes: The figure shows a binned scatterplot of the number of children of the 17,259 U.S. technologies with at least one parent, against their share of foreign parents. Birth-decade fixed effects are absorbed. Standard errors (in parentheses) are clustered by birth year. * $p < .10$, ** $p < .05$, *** $p < .01$.

the share of its own parents that are of foreign origin, absorbing birth-decade fixed effects. The relationship is strongly positive: within cohorts, a one percentage point increase in the foreign share of a technology's parents is associated with roughly 1.9% more children. The most fertile U.S. technologies also disproportionately build on foreign ideas.

One possible interpretation of this pattern echoes Granovetter's (1973) strength of weak ties: because domestic ideas diffuse more easily within the United States, most American inventors draw on much the same domestic knowledge, and the combinations it admits are more likely to have been tried. Foreign ideas are scarcer, and the recombinations they enable are more novel. We elaborate on this interpretation in our analysis below.

2 Ancestry and the Local Absorption of Foreign Technologies

2.1 The Local Stock of Ideas

The genealogy records which existing technologies inventors work with when founding new ones, and shows that foreign technologies figure prominently among them. A key question, then, is whether some U.S. locations are better at absorbing foreign ideas than others. To answer it, we must first measure the local idea stock.

For the purposes of our discussion, we say a technology is *absorbed* in a county once it is mentioned in patents filed by local inventors—direct evidence that inventors there have begun working with it. Formally, let $\mathcal{P}_{d,t}$ denote the set of patents filed in county d through year t , and $\mathcal{P}_{d,t}(\tau) = \mathcal{P}_{d,t} \cap \mathcal{P}(\tau)$ those among them that mention technology τ . The *local idea stock* is the set of technologies with which local inventors have begun to work,

$$\mathcal{A}_{d,t} = \{\tau : P_{d,t}(\tau) > 0\}, \quad (2)$$

where $P_{d,t}(\tau) = |\mathcal{P}_{d,t}(\tau)|$ is the number of local patents that mention τ . Membership in $\mathcal{A}_{d,t}$ requires only a single local patent; $P_{d,t}(\tau)$ records how intensively local inventors work with the technology through year t . The two objects thus characterize local absorption of technologies at the extensive and the intensive margin, respectively.

Because each technology has exactly one origin country, we write $\mathcal{A}_{d,t}^o$ for the technologies from origin o that county d has absorbed by t ; throughout, o indexes foreign countries. We then define

$$A_{o,d,t} = |\mathcal{A}_{d,t}^o| \quad (3)$$

for their number. Summing instead over patents,

$$A_{o,d,t}^P = \sum_{\tau \in \mathcal{A}_{d,t}^o} P_{d,t}(\tau) \quad (4)$$

weights each absorbed technology by the number of local patents that work with it: the intensive

margin of absorption at the level of the origin country. Throughout, a superscript P denotes the patent-weighted (intensive-margin) counterpart of an object. Finally, we write $a_d(\tau)$ for the year of county d 's first patent mentioning τ —the date at which d first absorbs τ , in analogy to its birth year $b(\tau)$.

Taken together, these objects provide a detailed characterization of the local stock of ideas that also allows us to measure empirical counterparts of a number of familiar objects from the theoretical literature. In particular, the number of technologies available in county d , $A_{d,t} = |\mathcal{A}_{d,t}|$, may be interpreted as an empirical counterpart of the idea stock A in [Romer \(1990\)](#) and [Aghion and Howitt \(1992\)](#). This local stock grows when local inventors create a new technology, when a technology is imported from a foreign country ([Grossman and Helpman, 1991](#); [Rivera-Batiz and Romer, 1991](#); [Eaton and Kortum, 1999](#); [Comin and Hobijn, 2010](#)), and when technologies diffuse from other U.S. locations ([Griliches, 1957](#)). We say a U.S.-origin technology is *invented* in county d if one of the founding patents is filed in county d ,

$$I_d(\tau) = \mathbf{1}\{\mathcal{P}_{d, b(\tau)+5} \cap \mathcal{F}(\tau) \neq \emptyset\}, \quad (5)$$

so that $I_d(\tau) = 1$ when d is among the technology's founding locations; otherwise, the technology reaches d , if at all, only after its founding is complete. Summing over the technologies a county absorbs in a given year, $I_{d,t} = \sum_{a_d(\tau)=t} I_d(\tau)$ counts the technologies invented in d at t . The remaining U.S.-origin technologies in the stock reached d from elsewhere in the United States; we write $\mathcal{A}_{d,t}^{US} = \{\tau \in \mathcal{A}_{d,t} : o(\tau) = US, I_d(\tau) = 0\}$ for this set and $A_{US,d,t}$ for its size. Because a technology is either invented in d or reaches it only after its founding is complete, the foreign, the invented, and the diffused parts of the stock are disjoint, and we can decompose the change of the local idea stock between $t - 1$ and t into three margins.

$$\Delta A_{d,t} = \underbrace{I_{d,t}}_{\text{1. locally invented}} + \underbrace{\sum_o \Delta A_{o,d,t}}_{\text{2. foreign origin}} + \underbrace{\Delta A_{US,d,t}}_{\text{3. U.S. origin}}. \quad (6)$$

[Kalyani et al. \(2025\)](#) study the first of these margins, documenting where new technologies are

pioneered within the United States and how the associated jobs then diffuse across regions. This paper turns to the second: the absorption of technologies from abroad, and its effect on the first, as counties that absorb foreign ideas go on to invent new technologies.

2.2 Ancestry and the Absorption of Foreign Ideas

Our hypothesis is that the stock of ideas from country o is larger where more residents trace their ancestry to o . Table 3 reports estimates of

$$\ln(1 + A_{o,d,2019}) = \delta_o + \delta_d + \beta \ln(1 + \text{Ancestry}_{o,d}^{2010}) + X'_{o,d} \zeta + \varepsilon_{o,d}, \quad (7)$$

where $\text{Ancestry}_{o,d}^{2010}$ is the number of residents of d , in thousands, who trace their ancestry to origin country o in the 2006–2010 American Community Survey (ACS), δ_o and δ_d are origin and destination fixed effects and $X_{o,d}$ contains cubic polynomials in the distance and in the difference in latitude between o and d .

A first finding is that the trade in ideas behaves much like the trade in goods: standard gravity terms—origin and destination fixed effects and distance controls (Head and Mayer, 2014)—account for about a third of the variation in the stock of absorbed foreign ideas ($R^2 = 0.32$). U.S. counties thus import more ideas from foreign countries that produce more of them, prolific importing counties import more ideas from everywhere, and idea imports fall with geographic distance.

Adding our measure of common ancestry raises the R^2 by 0.10, an increase of almost a third (column 1). Within the same origin and the same county, a 1% increase in ancestry from o is associated with 0.29% more technologies absorbed from o . This is in line with a large literature linking economic exchange across space to kinship and co-ethnic ties, including trade, foreign direct investment, and the diffusion of technology to migrants' home countries (Gould, 1994; Rauch and Trindade, 2002; Kerr, 2008). Social distance, as proxied by common ancestry, thus also appears to be central to the flow of ideas.

Of course, ancestry is not randomly assigned. A county whose industries need German machine tools may both import German technologies and attract German engineers, generating a

spurious correlation between ancestry and absorption. Column 3 therefore instruments ancestry with the push–pull instruments of [Burchardi et al. \(2019\)](#). The idea is to instrument historical migrations from a given origin country to a U.S. destination using the coincidence of timing: if large numbers of Germans came to the United States at a time when a U.S. county happened to be attracting migrants from all over the world, we expect many of its residents today to trace their ancestry there. Germans arrived mostly before 1900, when the Midwest was drawing migrants; Italians arrived mostly in 1900–1930, when the West and South were. The predicted distribution of ancestry thus reflects when each country’s migrants came and which counties within the United States were growing at the time, excluding any confounding factors that tie a specific origin country to a specific destination county.

This is what solves the problem. The push factor is measured from migrants who settled outside the county’s own census region, and the pull factor from migrants arriving from other continents, so that no feature of the origin–county pair enters the prediction itself. If German engineers settled in a U.S. county because its industries demanded German machine tools, the instrument would predict no German ancestry there at all. The same holds for any other factor specific to the origin–destination pair, whether it drives migration, technology imports, or both.⁴ The excluded instruments are vintages of these push–pull interactions, one for each of the ten census waves between 1880 and 2010 available in [Burchardi et al. \(2019\)](#), so that ancestry is predicted by the entire staggered history of migration episodes. Each regression uses only the vintages up to the year in which the endogenous variable is measured. Appendix Table [B.1](#) reports the first stage.

The estimated coefficient is, if anything, slightly larger (0.330, against 0.288 by OLS). Places with the right ethnic mix, therefore, hold a significant comparative advantage in absorbing ideas from the corresponding origin.

Column 4 repeats the estimation with an alternative measure of absorption that we use in the next section. *Non-generative* absorption relies only on patents that mention τ but none of τ ’s children: $P_{d,t}^{ng}(\tau)$ counts local patents that work with τ without mentioning any of its U.S.

⁴The same design has recently been used to study intergroup contact and prejudice ([Bursztyn et al., 2024](#)), the cultural origins of innovation ([Posch et al., 2026](#)), and the effect of immigration on local innovation and growth ([Terry et al., 2026](#)).

descendants, and $A_{o,d,t}^{ng}$ and $A_{o,d,t}^{P,ng}$ follow from it exactly as in equations (3) and (4). This is a demanding restriction, as it discards roughly half of all patents mentioning one of our technologies, but it leaves the results unchanged: the estimate is 0.317. Non-generative patents tend to record smaller advances within existing technologies, those that require no new terminology. A patent describing a process for producing polyvinyl chloride that is blue rather than white, for example, is non-generative. A patent that works with polyvinyl chloride in founding cling film, by contrast, is generative.⁵

Panel B repeats the same exercise at the intensive margin, replacing the count of absorbed technologies with the count of local patents that work with them, $\ln(1 + A_{o,d,2019}^P)$. The elasticities are larger (0.610 when instrumented, 0.499 by OLS). Ancestry predicts not only which foreign ideas reach a county but also how intensively local inventors work with them once they arrive.

Panel C repeats the same exercise with the flow of foreign technologies absorbed rather than the stock. We split the sample in 1980 and relate ancestry measured in the 1980 census to the flow of ideas absorbed thereafter, $\ln(1 + \Delta_{1980}^{2019} A_{o,d})$. The regressor now predates the outcome by construction. Ancestry stocks inherited from a century of migration predict which ideas a county goes on to absorb over the following four decades (0.455, s.e.=0.026, when instrumented). We use this timing repeatedly below.

3 The Diffusion of Foreign Technologies and U.S. Innovation

The previous section established that common ancestry gives some U.S. counties a comparative advantage in absorbing ideas from specific foreign origins. What is the causal effect of these foreign ideas on U.S. innovation? If new ideas are made from old ones, then U.S. counties that absorb more technologies from a given origin country should, over time, invent more technologies that build on the technologies from that origin. We start by showing this effect at the aggregate (origin country-destination county) level, before examining the more granular, technology-by-technology

⁵The distinction parallels one in the growth literature between innovations that improve existing products and innovations that create new ones (Acemoglu and Cao, 2015; Akcigit and Kerr, 2018; Garcia-Macia et al., 2019). Our unit is the technology rather than the firm's product: non-generative patents improve an existing technology, generative patents found a new one.

Table 3: The effect of co-ethnicity on foreign idea diffusion.

	(1)	(2)	(3)	(4)
	OLS	OLS	IV	IV
<i>Panel A. Stock of foreign ideas, extensive margin</i>				
	$\ln(1 + A_{o,d,2019})$		$\ln(1 + A_{o,d,2019}^{ng})$	
$\ln(1 + \text{Ancestry}_{o,d}^{2010})$	0.288*** (0.015)	0.288*** (0.015)	0.330*** (0.034)	0.317*** (0.034)
R ²	0.4191	0.4194		
Within R ²	0.1416	0.1419		
KP first-stage F			174.8	174.8
<i>Panel B. Stock of foreign ideas, intensive margin</i>				
	$\ln(1 + A_{o,d,2019}^P)$		$\ln(1 + A_{o,d,2019}^{P,ng})$	
$\ln(1 + \text{Ancestry}_{o,d}^{2010})$	0.499*** (0.025)	0.499*** (0.025)	0.610*** (0.064)	0.513*** (0.054)
R ²	0.3677	0.3680		
Within R ²	0.1338	0.1342		
KP first-stage F			174.8	174.8
<i>Panel C. Flow of foreign ideas, extensive margin</i>				
	$\ln(1 + \Delta_{1980}^{2019} A_{o,d})$		$\ln(1 + \Delta_{1980}^{2019} A_{o,d}^{ng})$	
$\ln(1 + \text{Ancestry}_{o,d}^{1980})$	0.328*** (0.016)	0.329*** (0.016)	0.455*** (0.026)	0.435*** (0.026)
R ²	0.3970	0.3971		
Within R ²	0.1582	0.1583		
KP first-stage F			97.1	97.1
Observations	606,213	606,213	606,213	606,213
Origin fixed effects	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes
Distance polynomials	No	Yes	Yes	Yes
Latitude polynomials	No	Yes	Yes	Yes

Notes: This table reports origin-country $\times$ destination-county regressions of technology absorption on ancestry. Panel A uses $\ln(1 + A_{o,d,2019})$, where $A_{o,d,2019}$ is the number of origin- o technologies absorbed by county d by 2019. Panel B uses $\ln(1 + A_{o,d,2019}^P)$, where $A_{o,d,2019}^P$ counts the local patents working with those technologies. Panel C uses $\ln(1 + \Delta_{1980}^{2019} A_{o,d})$, where $\Delta_{1980}^{2019} A_{o,d}$ is the number of origin- o technologies absorbed between 1980 and 2019. In column 4, absorption is computed using non-generative patents only. The independent variable is $\ln(1 + \text{Ancestry}_{o,d})$, where $\text{Ancestry}_{o,d}$ is the number of county- d residents, in thousands, reporting ancestry from origin o . Ancestry is measured in the 2006–2010 ACS in Panels A and B and in the 1980 census in Panel C. Column 1 reports OLS estimates, column 2 adds cubic controls for origin-destination distance and latitude difference, and columns 3 and 4 instrument ancestry using the push–pull census vintages available through the relevant ancestry year. All specifications include origin and destination fixed effects. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

variation in Section 5 below.

The key advantage of the genealogy is that it allows us to identify which U.S.-born technologies descend from which foreign parent technologies. Let C^o denote the set of technologies with at least one parent from origin o . The number of children of origin- o ideas invented in county d during a window $(t_0, t_1]$ is then

$$C_{o,d}^{t_0,t_1} = \#\{\tau' \in C^o : I_d(\tau') = 1 \text{ and } a_d(\tau') \in (t_0, t_1]\}, \quad (8)$$

where $a_d(\tau')$ is, as before, the year of county d 's first patent mentioning τ' . That is, $C_{o,d}^{t_0,t_1}$ counts the technologies that build on ideas from country o and that county d helped invent. Its patent-weighted counterpart $C_{o,d}^{P,t_0,t_1}$ counts the local patents mentioning them.⁶

Table 4 reports estimates from the following specification:

$$\ln(1 + C_{o,d}^{1980,2019}) = \delta_o + \delta_d + \beta \ln(1 + A_{o,d,1980}^{ng}) + X'_{o,d} \zeta + \varepsilon_{o,d}, \quad (9)$$

where the regressor is the county's stock of (non-generatively) absorbed origin- o ideas in 1980. Note this specification embodies the two precautions introduced above. First, the regressor shares no patents with the outcome and the patents that mention any children are excluded from the measure of absorption. Second, the regressor predates the outcome because the stock of foreign ideas is measured in 1980 while the outcome counts inventions post-1980.

Even so, the local absorption of foreign ideas is not randomly assigned, for the same reasons ancestry is not. For example, a county poised to invent plastics may absorb German chemistry precisely because its inventors intend to build on it. Columns 3 and 4 therefore instrument the absorption of foreign ideas with the same vintages of push-pull interactions as before. This isolates the variation in the 1980 stock of absorbed ideas that is driven purely by historical coincidence. Some U.S. counties happened to be booming, and so attracted migrants from *all* origins, precisely when German emigration peaked in the late nineteenth century. Other counties boomed decades

⁶A technology with parents from several origins appears in C^o for each of them, so these counts do not sum to the total number of children across origins. We show below that our findings hold under a unique mapping that assigns each child technology to a single origin: that of the foreign parent with the highest co-occurrence share $s_{\tau \rightarrow \tau'}$.

later, just as Italians began to arrive (Burchardi et al., 2019). It is this quasi-random component of historical settlement, rather than any factor specific to an origin–county pair, that identifies β .

The estimated elasticity in column 3 is 1.432 (s.e.=0.170), and the instruments are strong, with F statistics comfortably exceeding the relevant critical levels (Appendix Table C.1 reports the first stage). A U.S. county that increases its 1980 stock of absorbed ideas from a given origin by 1% invents 1.4% more of that origin’s child technologies over the following four decades. We can thus conclude that the local absorption of a foreign parent technology *causes* a comparative advantage in inventing its children. This is the paper’s central result: inventors do stand on the shoulders of giants, specifically on those that are locally available.

The identifying assumption is that, conditional on the origin and destination fixed effects and the geography controls, the interactions of migration pushes from origin o —measured from migrants who settled outside county d ’s census region—with county d ’s contemporaneous attractiveness to migrants from other continents are orthogonal to unobserved determinants of the invention of o ’s descendant technologies in d . That is, omitted factors that may make a particular U.S. county fertile ground for building on a particular origin country’s ideas must not also determine the historical interaction of nationwide migration waves from that country with the destination county’s general attractiveness to migrants from other continents at the time those waves arrived. The remaining threat is narrow: an unobserved characteristic of destinations in different census regions that is correlated with future invention and simultaneously caused large migrations from two origins on two different continents to settle in those same counties.

Robustness. Appendix C corroborates this identifying assumption with the established set of tests developed for this design (Burchardi et al., 2019; Bursztyn et al., 2024; Terry et al., 2026). For example, it shows that the estimate does not particularly hinge on any one wave of migration: instruments built only from the five census vintages of 1880–1930 give an elasticity of 1.34, and only from the 1970 and 1980 vintages 1.81, both significant at the 1 percent level (Appendix Table C.7). Moreover, results are robust to adding interacted fixed effects for each origin–census-region and each county–continent pair, which absorb any coincidence of settlement at the regional level

(Appendix Table C.4), and to controlling for the county’s 1930 and 1950 stocks of absorbed origin- o ideas, so that the estimate reflects absorption beyond what the county already held decades earlier (Appendix Table C.5). A placebo that assigns to each origin the child technologies of its alphabetical neighbor yields coefficients near zero, and the actual estimate exceeds all 1,000 random reassignments in a randomization test (Appendix Table C.8). No single origin country or innovation hub overturns the result: excluding Japan, the United Kingdom, Silicon Valley, the Northeast corridor, or the Chicago area one at a time leaves the elasticity between 1.35 and 1.41, and excluding Germany raises it to 2.81 (Appendix Table C.9).

The remaining columns and Panel B of Table 4 show plausible variations of how the two sides of this relationship are measured. Column 4 measures local innovation at the intensive margin, replacing the count of child technologies with the number of local patents that mention them, $\ln(1 + C_{o,d}^{P,1980,2019})$. The estimated coefficient implies that a 1% increase in the 1980 stock of absorbed ideas from a given origin raises subsequent local patenting on their descendants by 2.6% (coefficient 2.566). Panel B instead measures absorption at the intensive margin, weighting each absorbed technology by the number of local patents that mention it, $\ln(1 + A_{o,d,1980}^{P,ng})$. Because this regressor varies more across origin–county pairs than the count of technologies, the elasticities are naturally smaller (0.944 for the count of child technologies and 1.690 for the patents mentioning them), but they remain positive and statistically highly significant throughout. Whether absorption of foreign ideas and their descendants are better counted at the intensive or extensive margin, then, makes little difference to our conclusions. However we measure it, the local stock of absorbed foreign ideas is a strong determinant of subsequent local innovation building on them.

The baseline estimate in column 3 of Table 4 is also robust to how we set up the regression. The elasticity is 1.08 when each child technology is assigned to a single origin country, so that no child is counted twice (Appendix Table C.3). Similarly, it is robust to replacing the logarithm with the inverse hyperbolic sine on both sides and other plausible variations in the functional form (Appendix Table C.6). Appendix Table C.11 shows that the estimates remain significant at the 1 percent level if standard errors are clustered by county instead.

Finally, because the origin–destination panel is a balanced panel, it contains many countries

Table 4: The causal effect of foreign ideas on U.S. innovation.

	(1)	(2)	(3)	(4)
	OLS	OLS	IV	IV
<i>Panel A. Flow of US inventions with parents from origin o</i>				
	$\ln(1 + C_{o,d}^{1980,2019})$		$\ln(1 + C_{o,d}^{P,1980,2019})$	
$\ln(1 + A_{o,d,1980}^{ng})$	0.749*** (0.070)	0.749*** (0.070)	1.432*** (0.170)	2.566*** (0.291)
KP first-stage F			46.5	46.5
<i>Panel B. Flow of US inventions with parents from origin o</i>				
	$\ln(1 + C_{o,d}^{1980,2019})$		$\ln(1 + C_{o,d}^{P,1980,2019})$	
$\ln(1 + A_{o,d,1980}^{P,ng})$	0.533*** (0.047)	0.533*** (0.047)	0.944*** (0.107)	1.690*** (0.184)
KP first-stage F			76.5	76.5
Observations	606,213	606,213	606,213	606,213
Origin fixed effects	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes
Distance polynomials	No	Yes	Yes	Yes
Latitude polynomials	No	Yes	Yes	Yes
Anderson–Rubin F			74.1	63.1
Anderson–Rubin p			0.000	0.000
Stock-Yogo (10% size)			31.50	31.50
Stock-Yogo (5% rel. bias)			19.86	19.86

Notes: This table reports origin-country $\times$ destination-county regressions of U.S. child invention on the 1980 stock of absorbed foreign ideas. Panel A uses $\ln(1 + A_{o,d,1980}^{ng})$ as the independent variable, where $A_{o,d,1980}^{ng}$ is the number of origin- o technologies non-generatively absorbed by county d by 1980. Panel B uses $\ln(1 + A_{o,d,1980}^{P,ng})$, where $A_{o,d,1980}^{P,ng}$ counts the local non-generative patents working with those technologies. The dependent variable in columns 1–3 is $\ln(1 + C_{o,d}^{1980,2019})$, where $C_{o,d}^{1980,2019}$ is the number of U.S. child technologies with at least one parent of origin o whose first patent in county d falls in 1980–2019. In column 4 it is $\ln(1 + C_{o,d}^{P,1980,2019})$, where $C_{o,d}^{P,1980,2019}$ counts the local patents mentioning those child technologies. Column 1 reports OLS estimates, column 2 adds cubic controls for origin-destination distance and latitude difference, and columns 3 and 4 instrument absorption using the seven push–pull census vintages. All specifications include origin and destination fixed effects. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

from which no technology in our data originates. Exclusively estimating on the twenty origins that ever produce a U.S. child technology leaves the instruments strong, with first-stage F statistics between 59 and 87 on the restricted sample, and gives elasticities between 0.82 and 1.06 (Appendix Table C.10).

Mechanism. A key question now is whether the transmission of ideas works “on demand” through established kinship or other ties—transmitting ideas when they are needed from the origins that are socially proximate (Kerr, 2008, e.g.)—or whether inventors first need to work with the foreign technology locally before it can be recombined into something new (Arrow, 1962; Cohen and Levinthal, 1990, e.g.). Table 5 attempts to differentiate between these two views. Column 1 repeats the baseline from column 3 of Table 4. Columns 2–4 then add, one at a time, an alternative channel through which the historical settlement of migrants may transmit ideas: common ancestry in 1980, the foreign-born population from o in 1980, and the number of immigrants from o during the 1970s, each measured before 1980 like the absorption stock itself. Since these competing channels also reflect past immigration, we instrument them jointly with absorption. Because we have seven sets of instruments, one from each census vintage, we are able to differentially identify variation in non-generative absorption and the alternative measures of competing ties. The Sanderson–Windmeijer conditional F statistics confirm that the two endogenous variables are separately identified in each column.

All three competing channels enter positively, suggesting that co-ethnic ties transmit ideas through many margins. But the coefficient on non-generative absorption is remarkably stable, remaining between 1.10 and 1.39 throughout.⁷ On balance, when we compare estimated elasticities, ancestry ties alone appear to transmit relatively little. The lion’s share of the effect requires absorption — that local inventors first work with the foreign technology before it generates local descendants.

Columns 5 and 6 go one step further and add an indicator for FDI links with o in 2014 and

⁷Appendix Table C.2 repeats the horse race for the other specifications of Table 4, in which the outcome, the absorption stock, or both are weighted by patents. Adding a competing channel lowers the coefficient on absorption by at most a quarter (from 2.566 to between 1.957 and 2.499 when the outcome is weighted by patents), and it remains significant at the 1 percent level in every column.

Table 5: The causal effects of foreign ideas versus co-ethnicity.

	(1)	(2)	(3)	(4)	(5)	(6)
	IV	IV	IV	IV	IV	IV
<i>Flow US inventions with parents from origin o</i>						
	$\ln(1 + C_{o,d}^{1980,2019})$					
$\ln(1 + A_{o,d,1980}^{ng})$	1.432*** (0.170)	1.099*** (0.160)	1.166*** (0.174)	1.394*** (0.169)	1.121*** (0.165)	1.098*** (0.163)
$\ln(1 + \text{Ancestry}_{o,d}^{1980})$		0.198*** (0.023)				
$\ln(1 + \text{Foreign-born}_{o,d}^{1980})$			0.309*** (0.061)			
$\ln(1 + \text{Immigrants}_{o,d}^{1970-1980})$				0.179*** (0.049)		
FDI Dummy $_{o,d}^{2014}$					0.721*** (0.094)	
Information Demand Index $_{o,d}^{2016}$						0.316*** (0.071)
Observations	606,213	606,213	606,213	606,213	606,213	284,102
Origin fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes	Yes	Yes	Yes
Latitude polynomials	Yes	Yes	Yes	Yes	Yes	Yes
SW F (Non-Gen Abs)	46.5	65.1	80.8	64.2	76.0	47.3
SW F (Competitor)		86.8	434.6	626.2	74.2	30.1
Stock–Yogo (10% size)	31.50	29.18	29.18	29.18	29.18	29.18
Stock–Yogo (5% rel. bias)	19.86	19.28	19.28	19.28	19.28	19.28
Anderson–Rubin F	74.1	43.9	41.1	57.7	64.5	53.8
Anderson–Rubin p	0.000	0.000	0.000	0.000	0.000	0.000

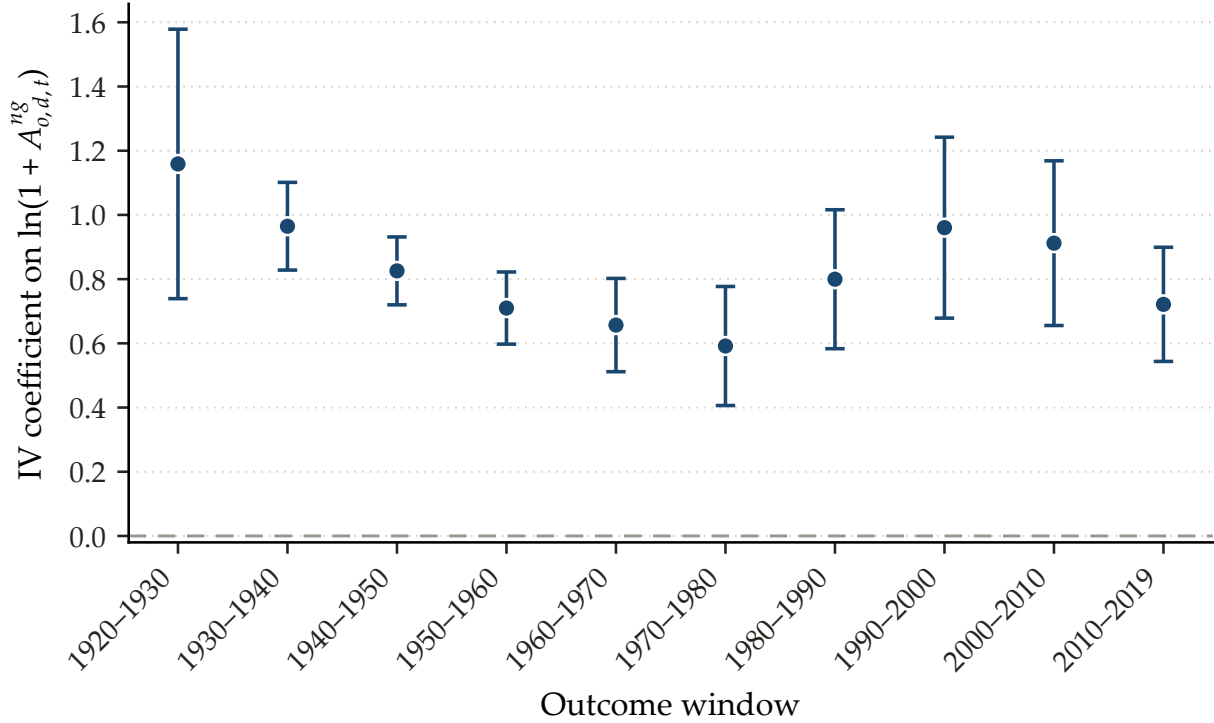
Notes: This table reports IV regressions of U.S. child invention on the 1980 stock of absorbed foreign ideas and one alternative channel at a time. The dependent variable is $\ln(1 + C_{o,d}^{1980,2019})$, the log number of U.S. child technologies with a parent of origin o first invented in county d in 1980–2019. The independent variable is $\ln(1 + A_{o,d,1980}^{ng})$, the county's 1980 stock of non-generatively absorbed origin- o technologies. Column 1 repeats column 3 of Table 4. Columns 2–4 add, one at a time, the log of one plus 1980 ancestry, the foreign-born population from o alive in 1980, and immigration from o during 1970–1980. Column 5 adds an indicator for FDI links with o in 2014 and column 6 an index of local demand for information about o in 2016. Each competitor is instrumented alongside absorption. The last two competitors are from Burchardi et al. (2019). Column 6's sample is smaller because the index is unavailable for counties outside a matched media market. All specifications include origin and destination fixed effects and cubic controls for origin-destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

an index of local demand for information about o in 2016. The first equals one if any firm in d has a parent or subsidiary in o , and the second measures how frequently residents of d search the internet for the names of o 's most prominent politicians, actors, athletes, and musicians.⁸ Because both variables are measured in the 2010s, they partly reflect consequences of the very idea flows we study. They may therefore absorb part of the true effect of absorption (they are “bad controls” in the sense of Angrist and Pischke (2009)). Even so, the coefficient on absorption stays in the same range as in columns 2–4 (1.12 and 1.10). The information-demand column is particularly interesting: in Burchardi et al. (2019), this index fully accounted for the effect of ancestry on FDI—suggesting that information flow was the fundamental mechanism that linked common ancestry to FDI outcomes. Here, the same variable moves the estimated effect of absorption no more than the other channels do. These patterns suggest that the recombination of ideas requires more than information about the origin country. In particular, it appears to require prior hands-on engagement with the technology itself.

Stability over time. Figure 4 asks whether this relationship is a feature of a particular era. It repeats the estimation of Table 4 column 3 decade by decade. The effect is remarkably stable: the elasticity is significantly positive in every decade and ranges between 0.6 and 1.2. The one systematic movement is a dip in the decades following World War II, when, as discussed in Section 1, foreign attribution at the USPTO was depressed, so that the stocks of foreign ideas measured in those years are the least complete. One might have expected the effect to fade in the most recent decades, as globalization and the internet made information about foreign technologies available everywhere at once. It does not do so significantly. Appendix Figure C.1 makes this comparison explicit by splitting the sample in 2000: the elasticity of child invention in 1980–2000 with respect to the 1980 stock (1.13) is statistically indistinguishable from that of invention in 2000–2019 with respect to the 2000 stock (1.05; $p = 0.141$ for the test of equality). Local absorption of a foreign idea has mattered for local invention building on it throughout the past century, and it still matters today.

⁸The FDI indicator is constructed from the 2014 edition of the ORBIS database, which lists the foreign parents and subsidiaries of each U.S. firm; the information-demand index averages the Google Trends popularity in d 's media market

Figure 4: The causal effect of foreign ideas on U.S. innovation, decade by decade.



Notes: The figure plots coefficient estimates with 95% confidence intervals from ten separate regressions, one for each decade of invention from 1920–1930 through 2010–2019. Each regression relates $\ln(1 + C_{o,d}^{t,t+10})$, the log number of U.S. child technologies with a parent of origin o first invented in county d during the decade, to $\ln(1 + A_{o,d,t}^{ng})$, the stock of non-generatively absorbed origin- o technologies at the decade's opening year t , in the specification of column 3 of Table 4. The confidence intervals use standard errors clustered by destination state.

4 The Diffusion of Foreign Technologies and U.S. Jobs

The previous two sections established that the local absorption of foreign ideas gives U.S. counties a comparative advantage in inventing the technologies that build on them. We next ask whether this advantage in innovation translates into local job opportunities.

To this end, we flag mentions of each technology in the full text of online job postings collected by Lightcast, which aggregates postings from online job boards and employer websites into a de-duplicated database recording the location, occupation, and full text of each posting, covering roughly 350 million postings from 2010 to 2023. We say a posting mentions a technology if its text contains one of the phrases associated with it, excluding the first and last 50 words of the posting, which typically describe the employer rather than the job. A human audit in [Kalyani et al. \(2025\)](#) shows that over 90% of such mentions refer to tasks that involve using the technology or supporting its production or operation, so that the number of local postings mentioning a technology is a direct measure of local jobs working with it.

For each origin country–U.S. county pair, we then count the postings in d over 2010–2023 that mention technologies of origin o , $A_{o,d}^J$, and those that mention U.S. child technologies with a parent of origin o , $C_{o,d}^J$, where the superscript J denotes an object counted in job postings. Because our interest is in the labor-market consequences of the absorption we measure in 1980, we restrict both counts to technologies born in 1980 or later, so that—by construction—neither the technology nor the job predates the regressor.

Table 6 reports estimates of the specification of column 3 of Table 4 with $\ln(1 + A_{o,d}^J)$ and $\ln(1 + C_{o,d}^J)$ as outcomes.⁹ Both effects are large and precisely estimated. A 1% increase in a county’s 1980 stock of (non-generatively) absorbed ideas from a given origin raises the number of local job postings mentioning technologies from that origin by 2.8% (s.e.=0.172), and the number mentioning U.S.-born child technologies by 3.5% (s.e.=0.254). We can thus conclude that absorbed foreign ideas shape the local labor market for long periods of time, both through jobs working with the imported technologies themselves and through jobs working with the domestic innovations

of the five most prominent names in each of the four categories. See [Burchardi et al. \(2019\)](#) for details on both variables.

⁹Because the regressor, the instruments, and the sample are those of Table 4, the first stage is the same (Appendix Table C.1).

they made possible.

Column 2 asks which jobs these are. Following [Kalyani et al. \(2025\)](#), we classify postings by the college-educated share of their six-digit SOC occupation: high-skill postings are those in occupations above 60 percent, low-skill postings those below 30 percent. A 1% increase in the 1980 stock raises high-skill postings by about 0.4% more than low-skill postings, whether the postings mention the origin’s own technologies (0.405, s.e.=0.034) or the U.S. technologies descended from them (0.409, s.e.=0.036). The jobs that absorbed ideas create are thus skewed toward high-skill occupations, but not exclusively so.

5 Local Idea Stocks and the Direction of Technical Change

5.1 Local Inputs and the Invention of Specific Technologies

The results so far concern aggregates: how many technologies a U.S. county invents that build on ideas from a given foreign country. We now sharpen the analysis to the level of the individual technology and ask not how much a U.S. county invents, but which *specific technologies* it invents. If new ideas are made from old ones, then a U.S. county should be more likely to invent a particular technology when that technology’s own parents are already in its idea stock. The genealogy lets us test this sharper implication directly: for every U.S.-origin technology τ' , we know its parents, and for every county d , we know which of them local inventors had absorbed by any given date.

Figures [5A](#) and [5B](#) show what this looks like for four major American technologies. Each row shows two maps of the same region: on the left, the counties in which all of the technology’s foreign parents had been absorbed before the technology was born; on the right, the counties that invented it, that is, mentioned the new technology in local patents within five years of its birth. If the availability of inputs shaped where a technology was invented, the second map should look like the first.

Polyvinylidene chloride, popularly known as cling film, was born in 1944. Four foreign parents – all of them German: butadiene (1913), polyvinyl acetate (1935), polyvinyl chloride (1936), and polystyrene (1937). The full German set had been absorbed prior to 1944 in only 17 U.S. counties, 16

Table 6: The causal effect of foreign ideas on U.S. job postings, high- versus low-skill.

	(1)	(2)
	IV	IV
	All	High – Low
<i>Panel A. Flow of job postings mentioning technologies from o</i>		
	$\ln(1 + A_{o,d}^I)$	
$\ln(1 + A_{o,d,1980}^{ng})$	2.759*** (0.172)	0.405*** (0.034)
KP first-stage F	46.5	46.5
Anderson–Rubin F	17.3	19.6
Anderson–Rubin p	0.000	0.000
<i>Panel B. Flow of job posting mentioning US inventions with parents from origin o</i>		
	$\ln(1 + C_{o,d}^I)$	
$\ln(1 + A_{o,d,1980}^{ng})$	3.473*** (0.254)	0.409*** (0.036)
KP first-stage F	46.5	46.5
Anderson–Rubin F	15.4	26.8
Anderson–Rubin p	0.000	0.000
Observations	606,213	606,213
Origin fixed effects	Yes	Yes
Destination fixed effects	Yes	Yes
Distance polynomials	Yes	Yes
Latitude polynomials	Yes	Yes
Stock-Yogo (10% size)	31.50	31.50
Stock-Yogo (5% rel. bias)	19.86	19.86

Notes: This table reports IV regressions of local job postings on the 1980 stock of absorbed foreign ideas. Panel A uses $\ln(1 + A_{o,d}^I)$, the log number of job postings in county d over 2010–2023 that mention a foreign technology of origin o . Panel B uses $\ln(1 + C_{o,d}^I)$, the postings that mention a U.S. child technology with a parent of origin o . Both count only technologies born in 1980 or later. The independent variable is $\ln(1 + A_{o,d,1980}^{ng})$, the county's 1980 stock of non-generatively absorbed origin- o technologies. Column 2 uses the difference between the two log outcomes obtained when the postings are split by the college-educated share of their six-digit SOC occupation, high-skill postings being those in occupations with a share above 60 percent and low-skill postings those with a share below 30 percent, and tests the equality of the two effects. Absorption is instrumented using the seven push–pull census vintages. All specifications include origin and destination fixed effects and cubic controls for origin-destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

of them in the East Coast window of Figure 5A. Founding patents for polyvinylidene chloride were filed in 28 counties, 22 of them in that region, and 10 of the 22 held the full set before 1944. They include the heartlands of the American chemical industry: Midland County, Michigan, where Dow first produced cling film under the commercial name of Saran, which completed the German set of parents in 1941; New Castle County, Delaware, home of DuPont, which had held all four since 1939 and had its first cling film patent in 1947; and Allegheny County, Pennsylvania, home of PPG, equipped since 1939 and inventing in 1949.

Photoresist, born in 1956, is photography before film: a coating that hardens where light strikes it, first used to take the earliest photograph and now to print circuits onto silicon. It has one foreign parent: polyvinyl alcohol (Germany, 1931), from which the first commercial photoresist was made. Polyvinyl alcohol had been absorbed in time in 142 U.S. counties, 94 of them in the East Coast window of Figure 5A. Of the 31 counties nationwide that invented photoresist within five years of 1956, 25 lie in that region and 18 absorbed the German input before 1956. Chief among them is Monroe County, New York, home of Eastman Kodak, whose Kodak Photo Resist opened the market: the county had held polyvinyl alcohol since 1929 and filed its first photoresist patent in 1947. Broome County, New York, was equipped in 1936 and patented in 1960; Schenectady County, New York, in 1950 (1960).

Evolution-Data Optimized (EV-DO, 2003) is the broadband network on which Alltel, Sprint and Verizon ran the early BlackBerry, Nokia, and Palm smartphones. One of its 38 parents is foreign: Bluetooth (Finland, 1999), which had been absorbed in time in 172 U.S. counties, 52 of them in the Midwest window of Figure 5B. EV-DO was invented in 59 counties, 15 of them in window, and 14 of the 15 held Bluetooth before 2003, from DuPage County, Illinois, home of Lucent's Bell Labs wireless laboratories, which absorbed Bluetooth in 2001 and filed the window's first EV-DO patent in 2002, to Cook and Lake Counties, Illinois, home of Motorola, and Johnson County, Kansas, home of Sprint, which absorbed Bluetooth in 2000 and patented on EV-DO in 2005.

Nanopore sequencing, born in 2008, reads a strand of DNA by threading it through a hole a few nanometres wide and recording the current across it. It has 99 parents, two of them foreign: the electron microscope (Germany, 1939), with which the pores are imaged, and the carbon nanotube

(Japan, 1995), the material of solid-state pores and of the electrodes that read them. Within the West Coast window (Figure 5B), the two had been absorbed in 25 counties before 2008, and all 9 counties that invented nanopore sequencing were among them (nationwide, 27 of 28). Santa Cruz County, California, home of the University of California laboratory that pioneered the method, had absorbed the carbon nanotube in 2004 and filed the window’s first nanopore-sequencing patent in 2005. The rest trace the genomics corridor of the Bay Area and beyond: San Mateo County, home of the Peninsula’s sequencing companies, had held both inputs since 1996; Santa Clara since 1986; San Francisco since 1999; San Diego County, home of Illumina, since 1999; and King County, Washington, since 2000.

In each case, the counties that invented the technology are, to a striking degree, the counties that already held its foreign inputs. To test whether the pattern holds systematically, we re-shape our dataset so that the unit of observation is a U.S.-origin technology τ' in a county d , and a five-year period t . Let $I_{d,t}(\tau') = \mathbf{1}\{a_d(\tau') \leq t\} \cdot I_d(\tau')$ indicate that county d has invented τ' by period t : it is zero until the period in which d first mentions τ' in a patent, and one thereafter if that patent falls within τ' ’s founding window (equation 5). Table 7 reports estimates of

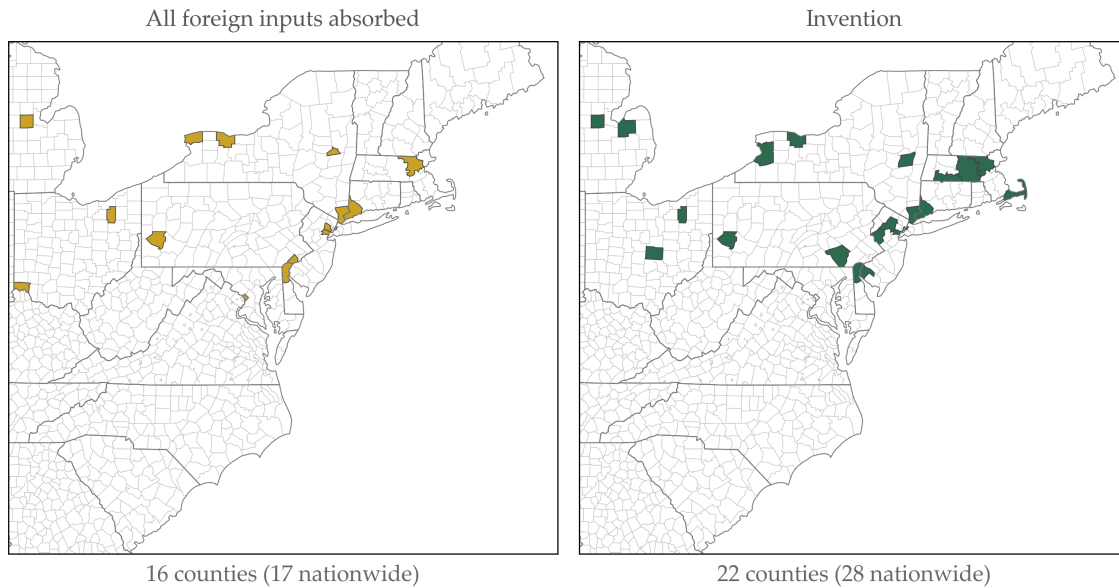
$$I_{d,t}(\tau') = \delta_o + \delta_d + \delta_t + \beta f_{\tau'}(\mathcal{A}_{d,t-1}) + X'_{o,d} \zeta + \varepsilon_{\tau',d,t}, \quad (10)$$

where $f_{\tau'}(\mathcal{A}_{d,t-1})$ summarizes which of τ' ’s parents were in county d ’s idea stock by the end of the previous period, $t - 1$, so that an input counts only if it was in place before the period in which the technology can be invented locally. The coefficient of interest, β , thus measures how much absorption of a technology’s specific inputs (parent technologies), on average, raises the probability that it is invented locally.¹⁰ The columns of the table differ only in $f_{\tau'}$. In column 1, it is an indicator that is one if all of τ' ’s parents are locally available; column 2 instead uses an indicator that is one if all foreign parents are available; columns 3 and 4 use an indicator that every foreign parent from the technology’s modal origin country—the country contributing the

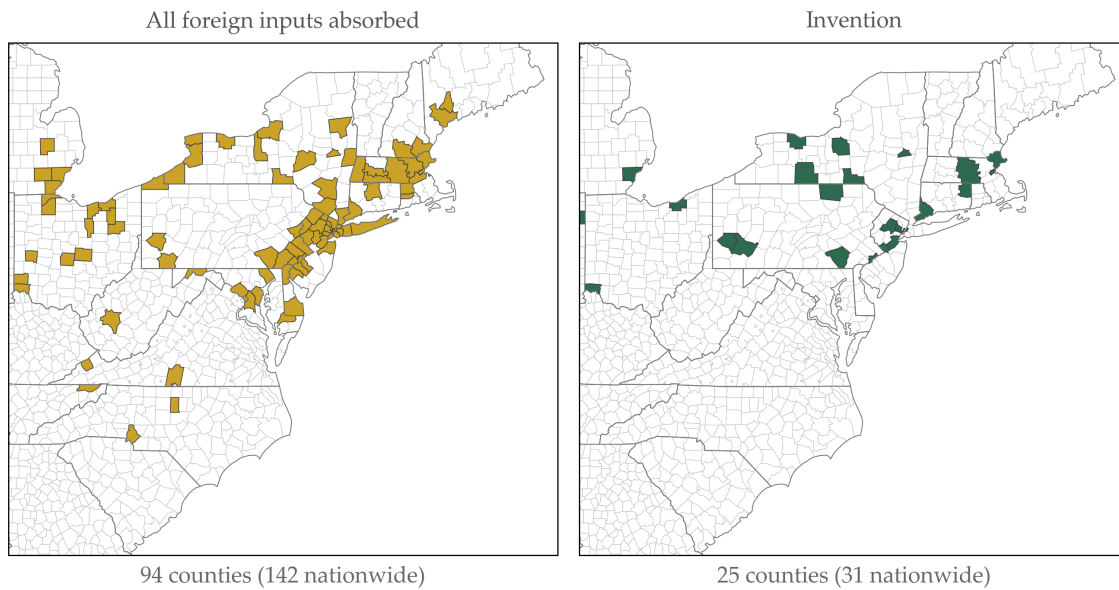
¹⁰In the language of the growth literature, equation (10) is a local, technology-level counterpart of the production function for ideas, in which the existing stock enters not as a scalar but as the specific set of inputs each new technology requires (Romer, 1990; Weitzman, 1998); in column 4, β is then the local average treatment effect of the presence of the specific inputs in $f_{\tau'}$ on the probability that τ' is invented locally.

Figure 5A: Foreign inputs and the invention of four U.S. technologies.

Polyvinylidene Chloride (Cling Film), 1944



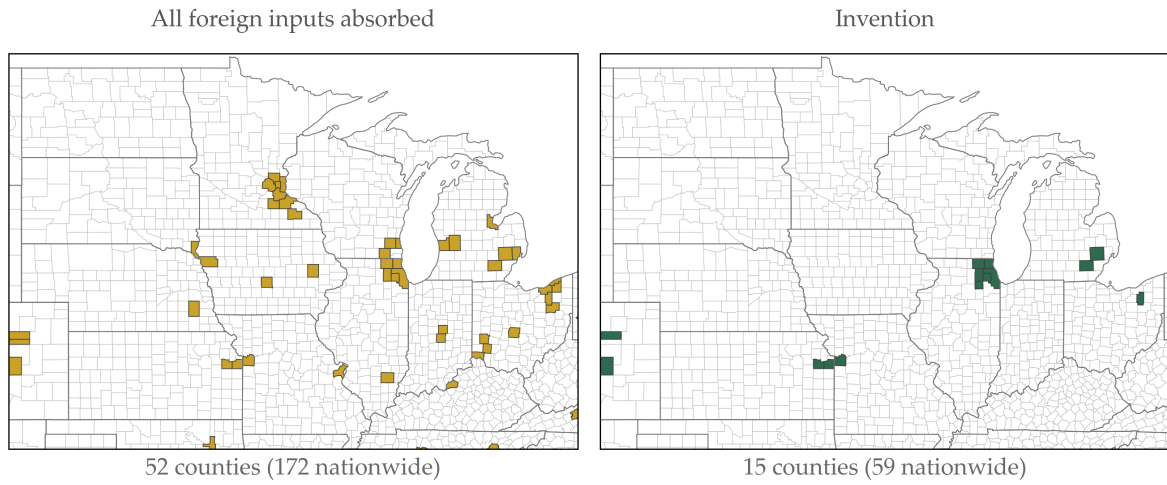
Photoresist, 1956



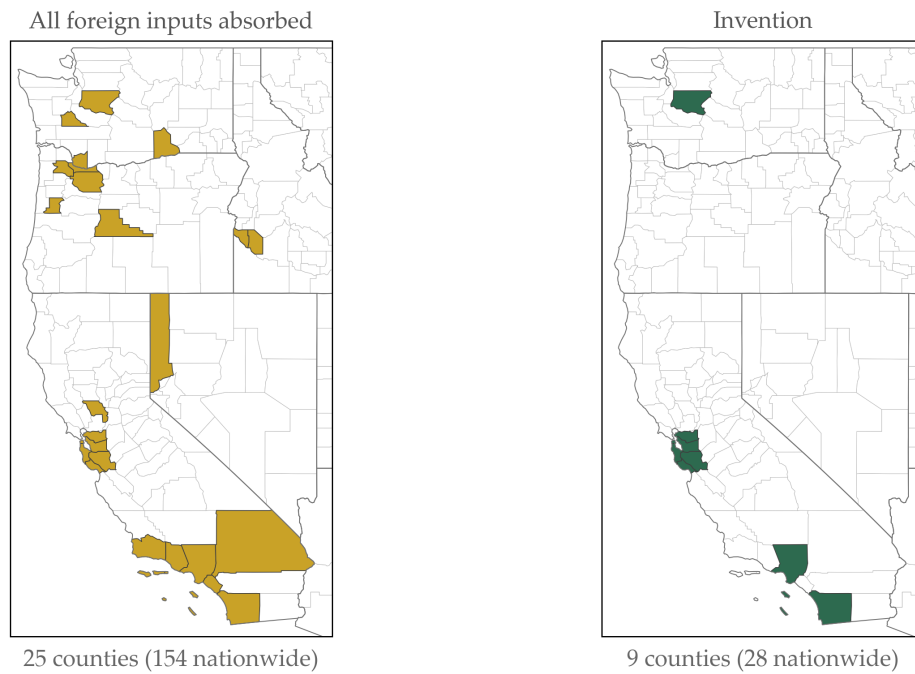
Notes: Each row takes one U.S. technology and shows two maps of the same region. The left map shows the counties in which all of the technology's foreign parents had been absorbed before the technology was born. The right map shows the counties that invented it, that is, patented on it within five years of its birth. This figure shows polyvinylidene chloride and photoresist, restricting to the East Coast of U.S..

Figure 5B: Foreign inputs and the invention of four U.S. technologies (continued).

EV-DO (Evolution-Data Optimized), 2003



Nanopore Sequencing, 2008



Notes: This page shows Evolution-Data Optimized in a Midwest window and nanopore sequencing, restricting to the West Coast of U.S.. The maps are constructed as in Figure 5A.

most of its foreign parents—is available.¹¹ δ_d , and δ_t are U.S. county, and period fixed effects, $X_{o,d}$ contains a cubic polynomial in distance to the modal parent country, and δ_o is a fixed effect for the model origin country o (where applicable). The latter two controls enter only in column 4, where availability is instrumented.

We limit the sample to U.S. technologies with more than ten children, which keeps the number of technology–county–period cells manageable, and to technologies with at least one foreign parent born after 1960, so that only post-1960 parents enter $f_{\tau'}$. The latter restriction focuses the analysis on the part of the sample in which foreign attribution is most accurate, although doing so has little influence on our results.

In column 1, U.S. counties in which all of a technology’s parents are locally available are significantly more likely to invent it (0.149, s.e.=0.013). Column 2 restricts the indicator to the technology’s foreign parents (0.055, s.e.=0.004). Column 3 restricts the kit to the foreign parents from the technology’s modal origin country (0.054, s.e.=0.004). Associating each technology with a single foreign origin country has a practical advantage: we can instrument for the availability of the parents from that country using the settlement history of its migrants, to which we turn next.

5.2 Causal Identification

The availability of a technology’s inputs is, of course, not randomly assigned across counties. A county whose inventors set out to build a particular new technology may import its foreign inputs for that express purpose, so that the local absorption of the parent is a consequence of the intent to invent the child rather than its cause. The estimates in columns 1–3 would then confound the role of inputs at hand.

We address this concern with a reshaped version of the same instruments as before (Burchardi et al., 2019; Terry et al., 2026). For each technology, we take the foreign country contributing the most of its foreign parents and instrument for the local availability of all of its parents from that foreign country with the push–pull interactions of that country’s migrants—one for each of the twenty-six five-year vintages between 1885 and 2010—together with fixed effects for that origin

¹¹Ties are broken in favor of the country with the highest total share, as defined in equation 1.

Table 7: The causal effect of foreign inputs on U.S. innovation.

	(1)	(2)	(3)	(4)
	OLS	OLS	OLS	IV
<i>US invention with foreign inputs</i> $I_{d,t}(\tau')$				
All inputs available at $t - 1$	0.149*** (0.013)			
All foreign inputs available at $t - 1$		0.055*** (0.004)		
All foreign inputs available at $t - 1$ (modal country)			0.054*** (0.004)	0.227*** (0.026)
Observations	19,214,020	19,214,020	19,214,020	19,214,020
R ²	0.1562	0.1546	0.1553	
Within R ²	0.0070	0.0050	0.0059	
Origin fixed effects	No	No	No	Yes
Destination fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Distance polynomials	No	No	No	Yes
KP first-stage F				95.4

Notes: This table reports linear-probability regressions of local child invention on the local availability of parent technologies, at the technology $\times$ county $\times$ five-year-period level. The sample comprises U.S.-origin technologies with more than ten children and at least one post-1960 foreign parent. The dependent variable equals one from the first period in which the county patents the technology within five years of its birth, and remains one thereafter. An input is locally available from the period of the county's first non-generative patent on it, and every indicator records availability at the end of the previous period. In column 1 the regressor is an indicator that all of the technology's post-1960 parents are available in the county. Column 2 uses the same indicator for its post-1960 foreign parents. Columns 3 and 4 use an indicator that every post-1960 foreign parent from the technology's modal origin country, the country contributing the most of them, is available. Columns 1–3 report OLS estimates and column 4 instruments the indicator using the twenty-six push-pull census vintages of the modal country, lagged one period. All specifications include county and period fixed effects, and column 4 adds modal-country fixed effects and a cubic control for distance to the modal country. Standard errors (in parentheses) are clustered by technology. * $p < .10$, ** $p < .05$, *** $p < .01$.

country. The identifying variation is again the coincidence of timing between a U.S. county's attractiveness to the average immigrant and the emigration waves from the parent technology's origin country, which—by construction—is orthogonal to any factor specific to the technology-county pair, including the intent to invent it.

Column 4 of Table 7 reports the result. The instruments are strong (Kleibergen–Paap F of 95.4), and the estimate is 0.227 (s.e.=0.026), about four times the least-squares estimate of column 3—consistent, as in Section 3, with significant measurement error in our text-based measure of absorption. The magnitude is large. Inventing any given technology in any given county is naturally a rare event: the dependent variable has a mean of 0.0079. Against this benchmark, local availability of a technology's foreign inputs from its modal origin country raises the probability that a county invents it by 22.7 percentage points—nearly thirty times the unconditional probability.¹²

The specification in column 4 requires a choice: which foreign parents' absorption to instrument for. Our baseline specification selects the origin country that contributes the most parents, and instruments for the availability of all of its parents. Table 8 varies that choice. Column 2 selects the origin country by the total dependence of the technology on its parents—the sum of their shares $s_{\tau \rightarrow \tau'}$ in the technology's founding patents (equation 1)—rather than by their number. Column 3 restricts the sample to technologies whose foreign parents all come from a single origin country, so that no choice is required. Column 4 instead instruments for the availability of the most recently born foreign parent alone. In each case we recover an estimate of β that is positive, of the same order, and statistically highly significant. What the maps show for cling film, photoresist, and wireless data networks thus holds at scale, and it holds causally: the U.S. counties that hold a foreign idea are the counties that go on to invent the specific technologies built on it. Local idea stocks thus determine not only how much a county invents, but what.

In this sense, our results speak to the direction of technical change. Since Hicks (1932), and in the modern literature since Acemoglu (1998, 2002), the direction of innovation has been understood from the demand side: inventors direct their efforts toward technologies with the largest markets or the scarcest substitutes. Our estimates identify a distinct force operating on the supply side.

¹²The estimate is smaller, 0.105, in a cross-section that keeps for each technology only the five-year period in which it was born (column 4 of Appendix Table D.1).

Table 8: The causal effect of foreign inputs on U.S. innovation, different inputs.

	(1)	(2)	(3)	(4)
	IV	IV	IV	IV
<i>US invention with foreign inputs</i> $I_{d,t}(\tau')$				
All foreign inputs available at $t - 1$ (modal country)	0.227*** (0.026)			
All foreign inputs available at $t - 1$ (top-share country)		0.181*** (0.024)		
All foreign inputs available at $t - 1$ (single country)			0.346*** (0.026)	
Most recent foreign parent available at $t - 1$				0.439*** (0.021)
Observations	19,214,020	19,214,020	11,346,710	19,214,020
Origin fixed effects	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes	Yes
KP first-stage F	95.4	128.8	72.7	88.2

Notes: This table reports the IV specification of column 4 of Table 7 under alternative rules for choosing the parents whose availability is instrumented. Column 1 repeats the baseline, column 4 of Table 7. Column 2 chooses instead the country whose post-1960 foreign parents have the largest total share. Column 3 keeps the technologies whose post-1960 foreign parents all come from one country (640 of the 1,250 technologies in the sample) and uses an indicator that the entire foreign kit is available. Column 4 uses an indicator that the most recently born post-1960 foreign parent is available. In each column the parent-country fixed effects, the distance control, and the push-pull instruments are those of the selected country, and the sample keeps the counties its push-pull series covers. Standard errors (in parentheses) are clustered by technology. * $p < .10$, ** $p < .05$, *** $p < .01$.

Robustness. Appendix D.1 reports the estimate under alternative definitions of the estimation sample. Because a county remains an inventor of a technology once it has patented on it, while the availability of the technology’s inputs keeps accumulating, the panel pairs invention with inputs that in some cells arrived after it. Restricting the panel to the periods in which the technology can still be invented gives 0.214. Additionally dropping the periods after a county has invented, so that the estimate is the chance of inventing among counties that have not yet done so, gives 0.132. Lastly, considering only the single period containing the technology’s birth year it is 0.105 (Appendix Table D.1). The effect is positive and significant at the 1 percent level throughout, and falls as the sample narrows toward the moment of invention. However, note that relative to each sample’s own invention rate the effect remains large. And these new estimates are the effect on inventing next, among counties that have not yet done so, against the baseline effect of having invented at all. Appendix Table D.2 shows that the estimate remains significant at the 1 percent level if standard errors are clustered by county or by destination state instead.

5.3 The Changing Geography of U.S. Innovation

If the local stock of foreign ideas determines what a county invents, then the geography of that stock should tell us something about the geography of American innovation, and changes in the stock about its changes. Our estimates absorb the total number of foreign ideas a U.S. county holds with county fixed effects, so that they identify the effect of foreign ideas from which origins’ ideas a county holds relative to other counties, never from how many. It is nevertheless instructive to see where foreign ideas have accumulated. Figure 6 maps each county’s stock of non-generatively absorbed foreign technologies in 2019, $\sum_o A_{o,d,2019}^{ng}$. These are the places that hold the most of the inputs from which new technologies are made. Foreign ideas are concentrated where one would expect: the Northeast corridor from Washington to Boston, the Great Lakes, coastal California and Puget Sound, and a set of metropolitan counties in the South and Mountain West.

What our estimates speak to directly are *changes* in this geography. Appendix Figures D.1 and D.2 decompose the 2019 stock into the part absorbed by 1980 and the part absorbed since. The 1980 stock is the geography of the manufacturing belt: Massachusetts, New York, New Jersey,

Pennsylvania, Ohio, Michigan, and Illinois, with California the only large concentration outside it. The 1980–2019 flow instead tilts toward the coasts and the Sunbelt: Southern California, Arizona, Texas, Florida, Georgia, and the Carolinas absorb a far larger share of the foreign ideas arriving.

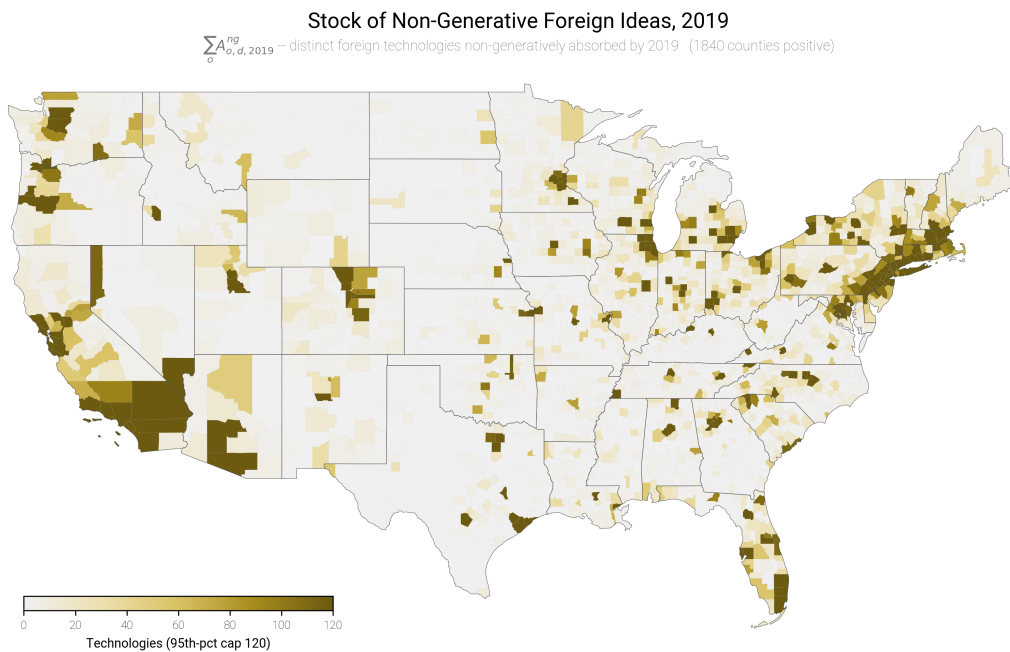
Our results imply that this shift has two sources, both of which are set in motion abroad. The ideas themselves have changed origin: as Table 1 shows, the most fertile foreign technologies of the early century were almost all German, those of mid-century most often British, while those of recent decades are most often Japanese, with Nordic technologies increasingly among them. And the settlement of migrants has changed with it: the descendants of the nineteenth-century migrations that gave the manufacturing belt its German and British ancestry did not settle where the post-1965 migrations from Asia and Latin America did. A county's comparative advantage in absorbing foreign ideas is therefore not fixed; it rises and falls with the match between the origins of its population and the origins of the ideas of the day.

Figure 7 shows how much of the recent reallocation of American innovation this mechanism accounts for. The left panel maps the change in each county's predicted rate of invention between the pre- and post-2000 periods, using only the fitted values from the instrumental-variable specification of column 4 of Table 7—that is, the change in the local availability of technologies' foreign inputs from their modal origin countries that is driven by the settlement history of those countries' migrants. The right panel maps the actual change in invention rates over the same periods. From both we subtract the national average in each period before taking the difference, so that the maps show which counties rose and which fell relative to the nation. Regressing the realized change on the predicted change across these counties (restricting to counties that ever invent a technology) gives a positive coefficient significant at the 1% and an R^2 of 0.21.

The losers are New York, New Jersey, Pennsylvania, and the industrial Midwest, whose stock of European ideas no longer matches the origins of the technologies being invented. The winners are Arizona, Florida, Massachusetts, Southern California, Texas, and the Southeast. For all but Massachusetts and Southern California, ancestry from the new source countries of ideas gives them access to the inputs of the technologies of the present. The changing origins of foreign ideas, and the changing map of who in America descends from where, is followed by the shift of

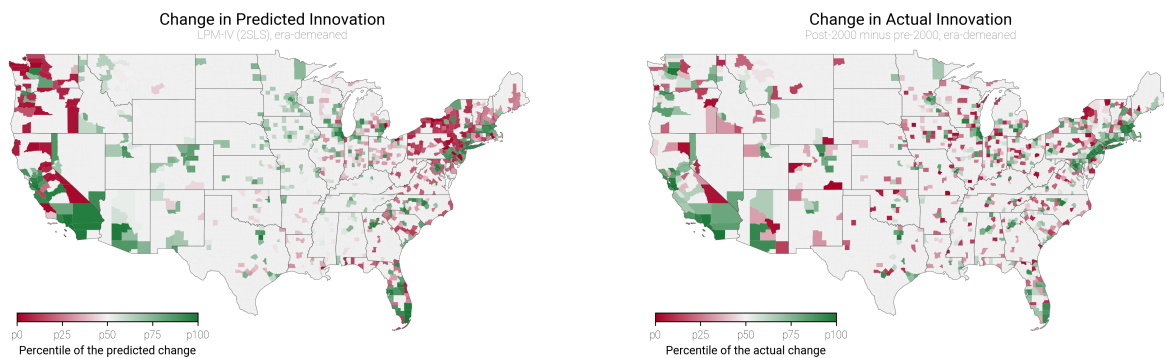
American innovation away from its historical heartland and toward the coasts and the Sunbelt.

Figure 6: Stock of non-generative foreign ideas, 2019.



Notes: Each county's stock of non-generatively absorbed foreign technologies in 2019, $\sum_o A_{o,d,2019}^{ng}$.

Figure 7: Change in predicted and realized innovation.



Ever-inventing counties, colored by within-panel percentile of the change (split at 2000; red = fell relative to the nation, green = rose; correlation 0.46, rank correlation 0.36)

Notes: The figure maps the post-2000 minus pre-2000 change in predicted and actual early-invention rates, with the national average of each period subtracted before the difference is taken. The predicted change uses the fitted value of the county fixed effect IV of column 4 of Table 7. Both panels are restricted to the counties that ever invent.

6 Conclusion

Key tenets in the study of innovation are that ideas build on ideas ([Schumpeter, 1911](#); [Romer, 1990](#)), and that ideas diffuse slowly across space ([Griliches, 1957](#); [Jaffe et al., 1993](#)). Together, they imply that the production of innovation has a geography: new ideas are made from old ones, and their manufacture depends on which ideas are locally at hand. Neither proposition has been easy to test directly, because two key data objects needed to test them—the production structure of ideas and the map of which ideas are available where—are typically treated as unobservable ([Krugman, 1991](#); [Ganguli et al., 2020](#)). Much of what we know about innovation is shaped by this constraint: growth theory treats the stock of ideas as a scalar; empirical work substitutes the similarity of patents for the dependence of one technology upon another; the direction of technical change is typically studied from the demand side alone; and the concentration of innovation in a handful of hubs is studied without observing the inputs from which those hubs make new ideas.

In this paper, we read both objects off the text of patents. We introduce a genealogy of technologies that records which specific earlier technologies were inputs into the creation of each new one. The location of the patents that mention each technology records when it entered each U.S. county’s stock of ideas. Because our unit of analysis is the technology rather than the patent, we can follow a technology from the country in which it was born, to the U.S. counties in which it was absorbed, to the technologies and jobs that were built from it.

We document four main findings. First, American innovation is built on foreign ideas: most U.S. technologies have at least one foreign parent technology, and the technologies with the most children disproportionately rely on foreign ideas as inputs.

Second, common ancestry gives U.S. counties a comparative advantage in absorbing ideas from specific foreign origins: a 1 percent increase in a U.S. county’s ancestry from an origin country raises its absorption of that origin’s technologies by about 0.46 percent. The ethnic composition of a county, inherited from a century of migration, is thus a channel through which the United States converts the world’s knowledge into innovation of its own—a return to immigration that operates over generations.

Third, this local absorption of ideas causes local invention. A 1% increase in a U.S. county’s

stock of absorbed ideas from a given foreign country raises its subsequent invention of technologies that build on technologies from that origin by 1.4%; and the channel appears to require hands-on experience with the foreign technology. These effects leave a lasting imprint in the local labor market: decades later, U.S. counties that absorbed more of an origin's ideas host more jobs working with those ideas, and more still working with the American technologies descended from them.

Fourth, at the level of the individual technology, U.S. counties invent the specific technologies whose inputs are already in their local idea stock: local availability of a technology's foreign inputs raises the probability that a county invents it many times over. The local stock of ideas thus determines not only how much a place invents but also *what* it invents. The post-2000 shift of American innovation from the manufacturing belt toward the coasts and the Sunbelt tracks the rise of Asia as a source of foreign ideas: as some of the most fertile foreign technologies came to originate in Japan rather than Germany, the comparative advantage in absorbing them moved to the U.S. locations whose populations include more descendants from Asia.

Our data thus make observable the stock and flow of technologies observable. The proposition that ideas build on ideas can be measured link by link, and the map of which ideas are available where can be drawn for every U.S. county and every year. This opens several avenues for future research.

First, our estimates are a reduced form of the production function for ideas. Incorporating the observed genealogy of technologies into models of idea production and diffusion ([Romer, 1990](#); [Buera and Oberfield, 2020](#)) would allow a structural estimate of that production function, at the cost of additional assumptions on how research effort is allocated across technologies. Such a model is also what is needed to move from allocation to aggregates. Our estimates identify how local access to foreign ideas allocates innovation across U.S. counties, but they do not speak directly to the aggregate gain to the United States from foreign ideas, which requires knowing how much invention in one county displaces invention elsewhere.

Second, our results complicate the conventional view that the local stock of ideas can be summarized by its size. Counties invent the *specific* technologies whose inputs they hold, so that the stock matters for its composition and not merely for its scale. The theory of directed technical

change has developed this idea on the demand side, where the direction of innovation follows market size and factor prices ([Acemoglu, 2002](#)). Our results suggest a supply-side counterpart: a theory in which the direction of innovation follows the composition of the ideas that inventors have to work with, and in which that composition differs across places.

Third, the genealogy provides data that network-based theories of recombinant growth require. These theories posit that new ideas arise from combinations of existing ones ([Weitzman, 1998](#); [Jones, 2023](#)), but their central object—which ideas combine with which, and how often—has not been observed. The genealogy makes it observable: how long a technology serves as an ingredient, how unequally that role is distributed across technologies, and how far the network of recombination is from the combinatorial explosion these models take as their engine.

Fourth, our results suggest that innovation hubs are places where the world’s ideas are gathered and recombined, and that which ideas gather where is in part the legacy of who settled there. Whether these links to the world are what make the United States the world’s innovation hub is a question our county-level estimates raise but cannot settle; recent cross-country evidence suggests that access to foreign ideas accounts for much of the American innovation lead over the twentieth century suggests it is worth pursuing ([Buera et al., 2026](#)).

Ideas build on ideas, and where they are built depends on which ideas are at hand. The objects we construct to show this—the stock of ideas, its flows, and the genealogy that links them—are defined by the patents that mention each technology, and can be measured for firms and sectors as readily as for U.S. counties. We hope that these data will prove useful to others in studying how technological innovations are made and how they travel.

References

- Acemoglu, D. (1998). Why do new technologies complement skills? directed technical change and wage inequality. Quarterly Journal of Economics 113(4), 1055–1089.
- Acemoglu, D. (2002). Directed technical change. Review of Economic Studies 69(4), 781–809.
- Acemoglu, D., U. Akcigit, and W. R. Kerr (2016). Innovation network. Proceedings of the National Academy of Sciences 113(41), 11483–11488.
- Acemoglu, D. and P. D. Azar (2020). Endogenous production networks. Econometrica 88(1), 33–82.
- Acemoglu, D. and D. Cao (2015). Innovation by entrants and incumbents. Journal of Economic Theory 157, 255–294.
- Acemoglu, D. and J. Linn (2004). Market size in innovation: Theory and evidence from the pharmaceutical industry. Quarterly Journal of Economics 119(3), 1049–1090.
- Acemoglu, D. and P. Restrepo (2022). Demographics and automation. Review of Economic Studies 89(1), 1–44.
- Aghion, P., A. Dechezleprêtre, D. Hémous, R. Martin, and J. Van Reenen (2016). Carbon taxes, path dependency, and directed technical change: Evidence from the auto industry. Journal of Political Economy 124(1), 1–51.
- Aghion, P. and P. Howitt (1992). A model of growth through creative destruction. Econometrica 60(2), 323–351.
- Ahmadpoor, M. and B. F. Jones (2017). The dual frontier: Patented inventions and prior scientific advance. Science 357(6351), 583–587.
- Akcigit, U. and W. R. Kerr (2018). Growth through heterogeneous innovations. Journal of Political Economy 126(4), 1374–1443.
- Angrist, J. D. and J.-S. Pischke (2009). Mostly Harmless Econometrics: An Empiricist’s Companion. Princeton, NJ: Princeton University Press.

- Arrow, K. J. (1962). The economic implications of learning by doing. Review of Economic Studies 29(3), 155–173.
- Bernstein, S., R. Diamond, A. Jiranaphawiboon, T. McQuade, and B. Pousada (2026). The contribution of high-skilled immigrants to innovation in the united states. American Economic Review. Forthcoming.
- Bloom, N., M. Schankerman, and J. Van Reenen (2013). Identifying technology spillovers and product market rivalry. Econometrica 81(4), 1347–1393.
- Buera, F. J., M. Kornejew, Y. Ma, E. Oberfield, and N. Trachter (2026). Innovation and development over the (very) long run. Working paper.
- Buera, F. J. and E. Oberfield (2020). The global diffusion of ideas. Econometrica 88(1), 83–114.
- Burchardi, K. B., T. Chaney, and T. A. Hassan (2019). Migrants, ancestors, and foreign investments. Review of Economic Studies 86(4), 1448–1486.
- Bursztyn, L., T. Chaney, T. A. Hassan, and A. Rao (2024). The immigrant next door. American Economic Review 114(2), 348–384.
- Cohen, W. M. and D. A. Levinthal (1990). Absorptive capacity: A new perspective on learning and innovation. Administrative Science Quarterly 35(1), 128–152.
- Comin, D. and B. Hobijn (2010). An exploration of technology diffusion. American economic review 100(5), 2031–2059.
- Davies, M. (2010). The corpus of historical american english (COHA). Available at <https://www.english-corpora.org/coha/>.
- Eaton, J. and S. Kortum (1999). International technology diffusion: Theory and measurement. International Economic Review 40(3), 537–570.
- Furman, J. L. and S. Stern (2011). Climbing atop the shoulders of giants: The impact of institutions on cumulative research. American Economic Review 101(5), 1933–1963.

- Galasso, A. and M. Schankerman (2015). Patents and cumulative innovation: Causal evidence from the courts. The Quarterly Journal of Economics 130(1), 317–369.
- Ganguli, I., J. Lin, and N. Reynolds (2020). The paper trail of knowledge spillovers: Evidence from patent interferences. American Economic Journal: Applied Economics 12(2), 278–302.
- Garcia-Macia, D., C.-T. Hsieh, and P. J. Klenow (2019). How destructive is innovation? Econometrica 87(5), 1507–1541.
- Gimbel, J. (1990). Science, Technology, and Reparations: Exploitation and Plunder in Postwar Germany. Stanford, CA: Stanford University Press.
- Gould, D. M. (1994). Immigrant links to the home country: Empirical implications for U.S. bilateral trade flows. Review of Economics and Statistics 76(2), 302–316.
- Granovetter, M. S. (1973). The strength of weak ties. American Journal of Sociology 78(6), 1360–1380.
- Griliches, Z. (1957). Hybrid corn: An exploration in the economics of technological change. Econometrica 25(4), 501–522.
- Gross, D. P. (2023). The hidden costs of securing innovation: The manifold impacts of compulsory invention secrecy. Management Science 69(4), 2318–2338.
- Grossman, G. M. and E. Helpman (1991). Innovation and Growth in the Global Economy. Cambridge, MA: MIT Press.
- Hall, B. H., A. B. Jaffe, and M. Trajtenberg (2001). The NBER patent citation data file: Lessons, insights and methodological tools. Working Paper 8498, National Bureau of Economic Research.
- Hanlon, W. W. (2015). Necessity is the mother of invention: Input supplies and directed technical change. Econometrica 83(1), 67–100.
- Hassan, T. A., A. Kalyani, and P. Restrepo (2026). The skill premium in times of rapid technological change. Working paper.

- Head, K. and T. Mayer (2014). Gravity equations: Workhorse, toolkit, and cookbook. In G. Gopinath, E. Helpman, and K. Rogoff (Eds.), Handbook of International Economics, Volume 4, pp. 131–195. Amsterdam: Elsevier.
- Hegde, D., K. Herkenhoff, and C. Zhu (2023). Patent publication and innovation. Journal of Political Economy 131(7), 1845–1903.
- Hicks, J. R. (1932). The Theory of Wages. London: Macmillan.
- Hunt, J. and M. Gauthier-Loiselle (2010). How much does immigration boost innovation? American Economic Journal: Macroeconomics 2(2), 31–56.
- Iaria, A., C. Schwarz, and F. Waldinger (2018). Frontier knowledge and scientific production: Evidence from the collapse of international science. The Quarterly Journal of Economics 133(2), 927–991.
- Jaffe, A. B. (1986). Technological opportunity and spillovers of R&D: Evidence from firms' patents, profits, and market value. American Economic Review 76(5), 984–1001.
- Jaffe, A. B., M. Trajtenberg, and R. Henderson (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. Quarterly Journal of Economics 108(3), 577–598.
- Jones, B. F. (2009). The burden of knowledge and the “death of the renaissance man”: Is innovation getting harder? Review of Economic Studies 76(1), 283–317.
- Jones, C. I. (2023). Recipes and economic growth: A combinatorial march down an exponential tail. Journal of Political Economy 131(8), 2382–2417.
- Kalyani, A., N. Bloom, M. Carvalho, T. A. Hassan, J. Lerner, and A. Tahoun (2025). The diffusion of new technologies. Quarterly Journal of Economics 140(2), 1299–1365.
- Keller, W. (2002). Geographic localization of international technology diffusion. American Economic Review 92(1), 120–142.
- Kelly, B., D. Papanikolaou, A. Seru, and M. Taddy (2021). Measuring technological innovation over the long run. American Economic Review: Insights 3(3), 303–320.

- Kerr, W. R. (2008). Ethnic scientific communities and international technology diffusion. Review of Economics and Statistics 90(3), 518–537.
- Kline, P., N. Petkova, H. Williams, and O. Zidar (2019). Who profits from patents? rent-sharing at innovative firms. Quarterly Journal of Economics 134(3), 1343–1404.
- Krugman, P. (1991). Geography and Trade. Cambridge, MA: MIT Press.
- Liu, E. and S. Ma (2026). Innovation networks and r&d allocation. American Economic Review. Forthcoming.
- Marx, M. and A. Fuegi (2020). Reliance on science: Worldwide front-page patent citations to scientific articles. Strategic Management Journal 41(9), 1572–1594.
- Moretti, E. (2010). Local multipliers. American Economic Review: Papers & Proceedings 100(2), 373–377.
- Moscona, J. and K. A. Sastry (2025). Inappropriate technology: Evidence from global agriculture. Working Paper 33500, National Bureau of Economic Research.
- Moser, P. and A. Voena (2012). Compulsory licensing: Evidence from the Trading with the Enemy Act. American Economic Review 102(1), 396–427.
- Moser, P., A. Voena, and F. Waldinger (2014). German jewish émigrés and US invention. American Economic Review 104(10), 3222–3255.
- Oberfield, E. (2018). A theory of input–output architecture. Econometrica 86(2), 559–589.
- Popp, D. (2002). Induced innovation and energy prices. American Economic Review 92(1), 160–180.
- Posch, M., J. F. Schulz, and J. Henrich (2026). How cultural diversity drives innovation: Surnames and patents in U.S. history. Journal of Political Economy 134(4), 1245–1305.
- Prato, M. (2025). The global race for talent: Brain drain, knowledge transfer, and growth. The Quarterly Journal of Economics 140(1), 165–238.

- Rauch, J. E. and V. Trindade (2002). Ethnic Chinese networks in international trade. Review of Economics and Statistics 84(1), 116–130.
- Rivera-Batiz, L. A. and P. M. Romer (1991). Economic integration and endogenous growth. Quarterly Journal of Economics 106(2), 531–555.
- Romer, P. M. (1990). Endogenous technological change. Journal of Political Economy 98(5), S71–S102.
- Sampat, B. and H. L. Williams (2019). How do patents affect follow-on innovation? evidence from the human genome. American Economic Review 109(1), 203–236.
- Schumpeter, J. A. (1911). The Theory of Economic Development. Cambridge, MA: Harvard University Press. English translation published 1934.
- Terry, S. J., T. Chaney, K. B. Burchardi, L. Tarquinio, and T. A. Hassan (2026). Immigration, innovation, and growth. American Economic Review 116(3), 828–861.
- Van Reenen, J. (1997). Employment and technological innovation: Evidence from U.K. manufacturing firms. Journal of Labor Economics 15(2), 255–284.
- Weitzman, M. L. (1998). Recombinant growth. Quarterly Journal of Economics 113(2), 331–360.
- Williams, H. L. (2013). Intellectual property rights and innovation: Evidence from the human genome. Journal of Political Economy 121(1), 1–27.
- Younge, K. A. and J. M. Kuhn (2016). Patent-to-patent similarity: A vector space model. SSRN Working Paper 2709238.

A Appendix to Section 1

A.1 Technology Explanations

We first restrict attention to Wikipedia pages with at least one section whose title contains one or more of the following terms: “application(s)”, “use(s)”, “type(s)”, “operation”, “characteristic(s)”, “feature(s)”, “device(s)”, “technical”, or “commercial”. We then classify the retained pages using a two-step prompt. The first step requires that a page describe a device or system of devices, a process, technique, or set of techniques, or a material designed for a specific purpose. Pages describing specific instances of technologies, such as a particular power plant, train model, or weapon model, or a specific company that makes devices, are excluded. The second step requires that the page describe at least one specific application. A page is classified as a technology only if it satisfies both steps. The full prompt is reproduced below.

You are a research assistant classifying Wikipedia pages.

Classify the page as technology or not_technology using this two-step test:

Step 1: Does the page describe one of these?

- a) A device or system of devices
- b) A process, technique, or set of techniques
- c) A material designed for a specific purpose

It must not describe a specific instance of any technology.

Eg: a specific power plant, a specific train model,
a specific weapon unit, a specific company that makes devices

If NO - classify as not_technology. It is likely one of:

- An abstract concept without a clear application
(e.g., frequency domain, band gap)
- An issue or problem topic
(e.g., disease progression, climate change)
- A management topic or business procedure
(e.g., business process)

- A person, place, event, organization, species,
or other non-technology topic
- A specific instance of a technology
(e.g., a particular power plant, a particular weapon model,
a particular company)

Step 2 (only if Step 1 = yes):

Does the page describe at least one specific application?

- If YES: classify as technology
- If NO: classify as not_technology
(e.g., a chemical compound with no described application)

Respond ONLY with a JSON object (no markdown, no commentary):

```
{
  "is_technology": true or false,
  "page_type": "device" | "technique" | "material" |
               "abstract_concept" | "problem_topic" |
               "management_topic" | "specific_instance" |
               "other_non_technology",
  "has_application": true or false,
  "classification_reason": "one sentence explanation",
  "confidence": "high" | "medium" | "low"
}
```

Table A.1: Examples from the Wikipedia technology-page classification.

Wikipedia page	LLM category	Classification rationale
<i>Panel A. Pages classified as technologies</i>		
3D printing	Technique	Applied manufacturing process for constructing three-dimensional objects.
CRISPR	Technique	Genome-editing technique with concrete medical, agricultural, and scientific applications.
Machine learning	Technique	Collection of algorithms and statistical techniques with applications across multiple domains.
Smartphone	Device	General class of computing and communication devices with numerous applications.
Jet engine	Device	General propulsion technology used in aviation and other applications.
Lithium-ion battery	Device	General energy-storage technology used in electronics, vehicles, and power systems.
Solar cell	Device	Device that converts light into electricity.
Semiconductor	Material	Material designed or modified for electronic applications.
<i>Panel B. Pages not classified as technologies</i>		
Quantum mechanics	Abstract concept	Scientific theory rather than an applied device, technique, or designed material.
Cancer	Problem topic	Disease that technologies may address, rather than a technology itself.
Digital divide	Problem topic	Socioeconomic problem concerning unequal access to technology.
Project management	Management topic	Organizational practice rather than a technological process.
Boeing 747	Specific instance	Particular aircraft model rather than a general technology class.
Ford Model T	Specific instance	Particular automobile model rather than a general technology class.
Microsoft	Specific instance	Particular company rather than a technology.

Notes: Panel A lists Wikipedia pages that the large language model classified as technologies and Panel B pages that it did not. The examples are drawn from the pages that passed the initial section-heading screen. The model classifies a page as a technology when it describes a generalizable device, technique, process, or designed material with at least one concrete application, and rejects abstract scientific concepts, problems, management practices, organizations, projects, and particular instances of a broader technology.

A.2 Technologies by Country of Origin

Table A.2: Technologies by country of origin.

Country	Technologies	Share
Japan	567	40.91
Germany	310	22.37
United Kingdom	157	11.33
France	59	4.26
Canada	53	3.82
Switzerland	45	3.25
South Korea	30	2.16
Sweden	20	1.44
Australia	19	1.37
Israel	18	1.30
Italy	18	1.30
China	15	1.08
Finland	14	1.01
Denmark	11	0.79
Netherlands	11	0.79
India	8	0.58
Belgium	6	0.43
Taiwan	4	0.29
Bulgaria	3	0.22
New Zealand	3	0.22
Norway	3	0.22
Spain	3	0.22
Others (≤ 2 each)	9	0.65
Total	1,386	100.00

Notes: Number and share of the 1,386 foreign-origin technologies by country of origin, the first country whose inventors file ten patents mentioning the technology. Others pools the countries with at most two technologies each.

B Appendix to Section 2

Table B.1: The effect of co-ethnicity on foreign idea diffusion, first stage.

	(1)	(2)
	OLS	OLS
	$\ln(1 + \text{Ancestry}_{o,d}^{2010})$	$\ln(1 + \text{Ancestry}_{o,d}^{1980})$
Push-pull, 1880	0.039*** (0.009)	0.038*** (0.009)
Push-pull, 1900	0.003 (0.015)	0.019 (0.019)
Push-pull, 1910	0.115*** (0.020)	0.116*** (0.019)
Push-pull, 1920	0.163*** (0.056)	0.156*** (0.047)
Push-pull, 1930	-0.171*** (0.066)	-0.198** (0.093)
Push-pull, 1970	0.126*** (0.035)	0.146*** (0.042)
Push-pull, 1980	0.051 (0.047)	0.066 (0.048)
Push-pull, 1990	0.046* (0.024)	
Push-pull, 2000	-0.223*** (0.036)	
Push-pull, 2010	0.337*** (0.043)	
Observations	606,213	606,213
Clusters (state)	51	51
Origin fixed effects	Yes	Yes
Destination fixed effects	Yes	Yes
Distance polynomials	Yes	Yes
Latitude polynomials	Yes	Yes
KP first-stage F	174.8	97.1
Stock-Yogo (10% size)	38.54	31.50
Stock-Yogo (5% rel. bias)	20.74	19.86

Notes: This table reports the first stages for the IV estimates of Table 3. Column 1 is the first stage for Panels A and B, with dependent variable $\ln(1 + \text{Ancestry}_{o,d}^{2010})$, where $\text{Ancestry}_{o,d}^{2010}$ is the number of county- d residents, in thousands, reporting ancestry from origin o in the 2006–2010 ACS. Column 2 is the first stage for Panel C, with dependent variable $\ln(1 + \text{Ancestry}_{o,d}^{1980})$, ancestry measured in the 1980 census. The 1990, 2000, and 2010 vintages do not enter and are left blank. Each row reports the coefficient on one push-pull census vintage. Both columns include origin and destination fixed effects and cubic controls for origin-destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

C Appendix to Sections 3 and 4

C.1 First Stage

Table C.1: The causal effect of foreign ideas on U.S. innovation, first stage.

	(1)	(2)
	OLS	OLS
	$\ln(1 + A_{o,d,1980}^{ng})$	$\ln(1 + A_{o,d,1980}^{P,ng})$
Push-pull, 1880	0.020*** (0.006)	0.027*** (0.008)
Push-pull, 1900	0.039*** (0.009)	0.065*** (0.012)
Push-pull, 1910	-0.015*** (0.004)	-0.029*** (0.006)
Push-pull, 1920	-0.012 (0.016)	-0.031 (0.026)
Push-pull, 1930	-0.039 (0.033)	-0.035 (0.049)
Push-pull, 1970	0.087*** (0.020)	0.131*** (0.030)
Push-pull, 1980	-0.033*** (0.010)	-0.053*** (0.016)
Observations	606,213	606,213
Clusters (state)	51	51
Origin fixed effects	Yes	Yes
Destination fixed effects	Yes	Yes
Distance polynomials	Yes	Yes
Latitude polynomials	Yes	Yes
KP first-stage F	46.5	76.5
Stock-Yogo (10% size)	31.50	31.50
Stock-Yogo (5% rel. bias)	19.86	19.86

Notes: This table reports the first stages for the IV estimates of Table 4. Column 1 is the first stage for Panel A, with dependent variable $\ln(1 + A_{o,d,1980}^{ng})$, the number of origin- o technologies non-generatively absorbed by county d by 1980. Column 2 is the first stage for Panel B, with dependent variable $\ln(1 + A_{o,d,1980}^{P,ng})$, the count of local non-generative patents working with those technologies by 1980. Each row reports the coefficient on one push-pull census vintage. Both columns include origin and destination fixed effects and cubic controls for origin-destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

C.2 Alternative Channels

Table C.2: The causal effect of foreign ideas versus co-ethnicity, under alternative specifications.

	(1)	(2)	(3)	(4)
	IV	IV	IV	IV
<i>Panel A. Patent-weighted outcome (Table 4, Panel A, column 4)</i>				
	$\ln(1 + C_{o,d}^{P,1980,2019})$			
$\ln(1 + A_{o,d,1980}^{ng})$	2.566*** (0.291)	1.957*** (0.272)	2.090*** (0.298)	2.499*** (0.289)
$\ln(1 + \text{Ancestry}_{o,d}^{1980})$		0.364*** (0.040)		
$\ln(1 + \text{Foreign-born}_{o,d}^{1980})$			0.555*** (0.111)	
$\ln(1 + \text{Immigrants}_{o,d}^{1970-1980})$				0.323*** (0.091)
SW F (Non-Gen Abs)	46.5	65.1	80.8	64.2
SW F (Competitor)		86.8	434.6	626.2
<i>Panel B. Patent-weighted absorption (Table 4, Panel B, column 3)</i>				
	$\ln(1 + C_{o,d}^{P,1980,2019})$			
$\ln(1 + A_{o,d,1980}^{P,ng})$	0.944*** (0.107)	0.726*** (0.098)	0.771*** (0.108)	0.918*** (0.106)
$\ln(1 + \text{Ancestry}_{o,d}^{1980})$		0.204*** (0.021)		
$\ln(1 + \text{Foreign-born}_{o,d}^{1980})$			0.311*** (0.060)	
$\ln(1 + \text{Immigrants}_{o,d}^{1970-1980})$				0.197*** (0.049)
SW F (Non-Gen Abs)	76.5	89.9	108.0	84.7
SW F (Competitor)		104.8	492.2	649.1
<i>Panel C. Both patent-weighted (Table 4, Panel B, column 4)</i>				
	$\ln(1 + C_{o,d}^{P,1980,2019})$			
$\ln(1 + A_{o,d,1980}^{P,ng})$	1.690*** (0.184)	1.288*** (0.168)	1.379*** (0.186)	1.643*** (0.183)
$\ln(1 + \text{Ancestry}_{o,d}^{1980})$		0.376*** (0.037)		
$\ln(1 + \text{Foreign-born}_{o,d}^{1980})$			0.561*** (0.109)	
$\ln(1 + \text{Immigrants}_{o,d}^{1970-1980})$				0.355*** (0.092)
SW F (Non-Gen Abs)	76.5	89.9	108.0	84.7
SW F (Competitor)		104.8	492.2	649.1
Observations	606,213	606,213	606,213	606,213
Origin fixed effects	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes	Yes
Latitude polynomials	Yes	Yes	Yes	Yes

Notes: This table repeats the horse race of Table 5 under the remaining specifications of Table 4. Panel A uses the patent-weighted outcome $\ln(1 + C_{o,d}^{P,1980,2019})$ with the extensive-margin stock $\ln(1 + A_{o,d,1980}^{ng})$. Panel B uses the child-technology count $\ln(1 + C_{o,d}^{1980,2019})$ with the patent-weighted stock $\ln(1 + A_{o,d,1980}^{P,ng})$. Panel C uses the patent-weighted outcome with the patent-weighted stock. Column 1 of each panel repeats the corresponding column of Table 4. Columns 2–4 add the log of one plus 1980 ancestry, the 1980 foreign-born stock, and immigration during 1970–80, each instrumented alongside absorption using the seven push–pull census vintages. All specifications include origin and destination fixed effects and cubic controls for origin-destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

C.3 Alternative Specifications

Table C.3: The causal effect of foreign ideas on U.S. innovation, unique-origin assignment of child technologies.

	(1)	(2)	(3)	(4)
	OLS	OLS	IV	IV
<i>Panel A. Flow of US inventions with parents from origin o</i>				
	$\ln(1 + \tilde{C}_{o,d}^{1980,2019})$		$\ln(1 + \tilde{C}_{o,d}^{P,1980,2019})$	
$\ln(1 + A_{o,d,1980}^{ng})$	0.520*** (0.054)	0.520*** (0.054)	1.084*** (0.148)	2.133*** (0.271)
KP first-stage F			46.5	46.5
Anderson–Rubin F			114.7	91.1
Anderson–Rubin p			0.000	0.000
<i>Panel B. Flow of US inventions with parents from origin o</i>				
	$\ln(1 + \tilde{C}_{o,d}^{P,1980,2019})$		$\ln(1 + \tilde{C}_{o,d}^{P,1980,2019})$	
$\ln(1 + A_{o,d,1980}^{P,ng})$	0.360*** (0.036)	0.360*** (0.036)	0.713*** (0.093)	1.403*** (0.171)
KP first-stage F			76.5	76.5
Anderson–Rubin F			114.7	91.1
Anderson–Rubin p			0.000	0.000
Observations	606,213	606,213	606,213	606,213
Origin fixed effects	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes
Distance polynomials	No	Yes	Yes	Yes
Latitude polynomials	No	Yes	Yes	Yes
Stock-Yogo (10% size)			31.50	31.50
Stock-Yogo (5% rel. bias)			19.86	19.86

Notes: This table repeats Table 4 with each child technology assigned to a single origin. Panel A uses $\ln(1 + A_{o,d,1980}^{ng})$ as the independent variable and Panel B uses $\ln(1 + A_{o,d,1980}^{P,ng})$. The dependent variable in columns 1–3 is $\ln(1 + \tilde{C}_{o,d}^{1980,2019})$, where $\tilde{C}_{o,d}^{1980,2019}$ counts the U.S. child technologies whose first patent in county d falls in 1980–2019 and whose assigned origin is o , the origin of the child’s foreign parent with the highest co-occurrence share, with ties broken by the earlier-born parent. In column 4 it is the patent-weighted counterpart $\ln(1 + \tilde{C}_{o,d}^{P,1980,2019})$. Column 1 reports OLS estimates, column 2 adds cubic controls for origin-destination distance and latitude difference, and columns 3 and 4 instrument absorption using the seven push–pull census vintages. All specifications include origin and destination fixed effects. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C.4: The causal effect of foreign ideas on U.S. innovation, fixed-effect ladder.

	(1)	(2)	(3)
	IV	IV	IV
	$\ln(1 + C_{o,d}^{1980,2019})$		
$\ln(1 + A_{o,d,1980}^{ng})$	1.432*** (0.170)	1.762*** (0.135)	1.571*** (0.116)
Observations	606,213	606,213	606,213
Origin fixed effects	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes
Latitude polynomials	Yes	Yes	Yes
Origin $\times$ region fixed effects	No	Yes	Yes
Destination $\times$ continent fixed effects	No	No	Yes
KP first-stage F	46.5	32.8	69.7
Stock-Yogo (10% size)	31.50	31.50	31.50
Stock-Yogo (5% rel. bias)	19.86	19.86	19.86
Anderson–Rubin F	74.1	40.9	67.1
Anderson–Rubin p	0.000	0.000	0.000

Notes: This table adds finer geographic fixed effects to the baseline. Column 1 repeats column 3 of Table 4. Column 2 adds origin $\times$ destination-region fixed effects, and column 3 in addition interacts the destination fixed effects with the origin's continent. All specifications instrument absorption using the seven push–pull census vintages and include cubic controls for origin–destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C.5: The causal effect of foreign ideas on U.S. innovation, pre-period stocks.

	(1)	(2)	(3)
	IV	IV	IV
	$\ln(1 + C_{o,d}^{1980,2019})$		
$\ln(1 + A_{o,d,1980}^{ng})$	1.432*** (0.170)	1.769*** (0.378)	2.886** (1.460)
$\ln(1 + A_{o,d,1920}^{ng})$		-1.528** (0.763)	
$\ln(1 + A_{o,d,1950}^{ng})$			-2.684 (1.829)
KP first-stage F	46.5	42.8	46.8
Observations	606,213	606,213	606,213
Origin fixed effects	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes
Latitude polynomials	Yes	Yes	Yes

Notes: This table controls for the county's earlier absorption. Column 1 repeats column 3 of Table 4. Columns 2 and 3 add the county's 1920 and 1950 stocks of non-generatively absorbed origin- o technologies, $\ln(1 + A_{o,d,1920}^{ng})$ and $\ln(1 + A_{o,d,1950}^{ng})$. All specifications instrument the 1980 stock using the seven push-pull census vintages and include origin and destination fixed effects and cubic controls for origin-destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C.6: The causal effect of foreign ideas on U.S. innovation, functional form.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	IV	IV	IV	IV	IV	IV	IV
	Log / Log	Log / IHS	Log / Level	IHS / Log	Level / Log	IHS / IHS	Level / Level
$\ln(1 + A_{o,d,1980}^{ng})$	1.432*** (0.170)			1.659*** (0.185)	59.312*** (16.734)		
IHS $A_{o,d,1980}^{ng}$		1.207*** (0.145)				1.398*** (0.158)	
$A_{o,d,1980}^{ng}$ (level)			0.108*** (0.012)				4.552*** (1.148)
Observations	606,213	606,213	606,213	606,213	606,213	606,213	606,213
KP first-stage F	46.5	42.1	146.1	46.5	46.5	42.1	146.1
Stock-Yogo (10% size)	31.50	31.50	31.50	31.50	31.50	31.50	31.50
Stock-Yogo (5% rel. bias)	19.86	19.86	19.86	19.86	19.86	19.86	19.86

Notes: This table repeats column 3 of Table 4 under alternative transformations of the outcome and the endogenous variable, given in the column heads as outcome / endogenous variable. Log is $\ln(1 + x)$, IHS the inverse hyperbolic sine, and Level the raw count, the number of U.S. child technologies invented in 1980–2019 and the number of origin- o technologies non-generatively absorbed by 1980. All specifications instrument absorption using the seven push-pull census vintages and include origin and destination fixed effects and cubic controls for origin-destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

C.4 Validity of the Instruments

Table C.7: The causal effect of foreign ideas on U.S. innovation, instrument vintages.

	(1)	(2)	(3)
	IV	IV	IV
	All seven vintages	1880–1930 only	1970–1980 only
	$\ln(1 + C_{o,d}^{1980,2019})$		
$\ln(1 + A_{o,d,1980}^{ng})$	1.432*** (0.170)	1.338*** (0.150)	1.807*** (0.195)
KP first-stage F	46.5	44.1	38.7
Observations	606,213	606,213	606,213
Excluded instruments	7	5	2
Equality of (2) and (3), p		0.002	
Origin fixed effects	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes
Latitude polynomials	Yes	Yes	Yes

Notes: This table repeats column 3 of Table 4 with subsets of the push–pull census vintages as instruments. Column 1 uses all seven, column 2 the five 1880–1930 vintages, and column 3 the 1970 and 1980 vintages. The p -value tests the equality of the column 2 and 3 estimates in a stacked estimation of the two columns. All specifications include origin and destination fixed effects and cubic controls for origin–destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C.8 probes whether the push–pull instruments isolate origin-specific settlement with a placebo in the spirit of Burchardi et al. (2019), reassigning outcomes across origins alphabetically. Randomization inference makes the reassignment formal: we permute the dependent variable across origins at random—each origin keeping its own absorption stock and its own instruments—and re-estimate the specification of column 3 of Table 4 on each of 1,000 such permutations. If the instruments carried generic county-level variation that attracts invention from every origin, reassigned outcomes would respond to them as well. They do not: across the 1,000 draws the placebo coefficient has a median of -0.07 and never exceeds 0.88 in absolute value, so the actual estimate of 1.432 lies outside the entire permutation distribution, a randomization p -value of 0.001 . Because this p -value comes from the permutation distribution itself, it requires no assumptions

about the structure of the errors, only that origins are exchangeable under the null.

Table C.8: The causal effect of foreign ideas on U.S. innovation, alphabetical-neighbor placebo.

	(1)	(2)	(3)
	IV	IV	IV
	$\ln(1 + C_{o,d}^{1980,2019})$	$\ln(1 + C_{n(o),d}^{1980,2019})$	$\ln(1 + C_{n'(o),d}^{1980,2019})$
$\ln(1 + A_{o,d,1980}^{ng})$	1.432*** (0.170)	-0.043** (0.018)	-0.030* (0.018)
KP first-stage F	46.5	46.5	46.5
Observations	606,213	606,213	606,213
Origin fixed effects	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes
Latitude polynomials	Yes	Yes	Yes

Notes: This table repeats column 3 of Table 4 with the dependent variable replaced by the child count of another origin. Column 2 uses the alphabetically adjacent origin $n(o)$ and column 3 the nearest alphabetical origin on a different continent $n'(o)$, which rules out neighbors that share a region (Niger and Nigeria, Slovakia and Slovenia). Origin o keeps its own absorption stock and instruments. All specifications include origin and destination fixed effects and cubic controls for origin-destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

C.5 Influential Origins and Places

Table C.9: The causal effect of foreign ideas on U.S. innovation, excluding origins and destination.

	(1) IV	(2) IV	(3) IV	(4) IV	(5) IV	(6) IV	(7) IV
	Baseline	excl. Germany	excl. Japan	excl. UK	excl. Silicon Valley	excl. Northeast corridor	excl. Chicago area
	$\ln(1 + C_{o,d}^{1980,2019})$						
$\ln(1 + A_{o,d,1980}^{1980})$	1.432*** (0.170)	2.811*** (0.316)	1.412*** (0.169)	1.353*** (0.166)	1.354*** (0.144)	1.361*** (0.214)	1.413*** (0.179)
KP first-stage F	46.5	58.7	42.6	36.1	39.9	48.6	43.9
Observations	606,213	603,072	603,072	603,072	605,441	598,686	604,862
Origin fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Latitude polynomials	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table repeats column 3 of Table 4 excluding, one at a time, the three largest origin countries of foreign technologies and three destination clusters. The clusters are Silicon Valley (Alameda, San Francisco, San Mateo, and Santa Clara counties), the Northeast corridor (the urban counties of the Boston, Providence, Hartford, New York, Philadelphia, Baltimore, and Washington areas), and the Chicago area (Cook, DuPage, Kane, Lake, McHenry, and Will counties in Illinois, and Lake County, Indiana). All specifications instrument absorption using the seven push-pull census vintages and include origin and destination fixed effects and cubic controls for origin-destination distance and latitude difference. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

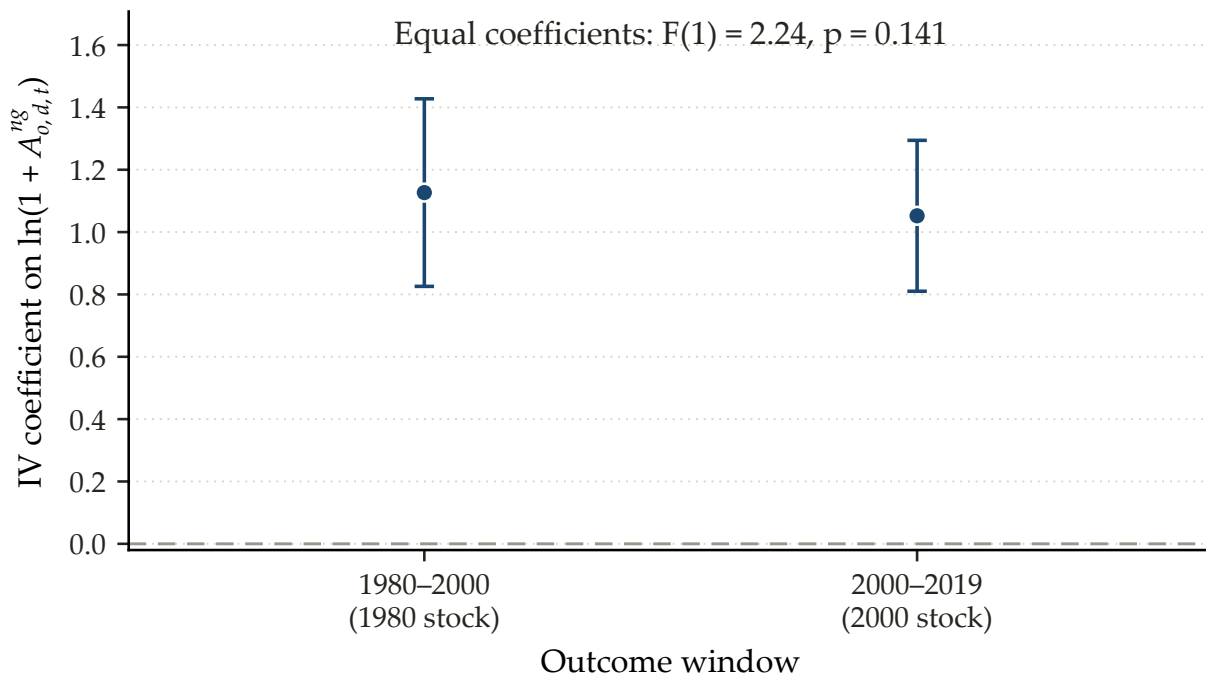
Table C.10: The causal effect of foreign ideas on U.S. innovation, origins with child technologies.

	(1)	(2)	(3)	(4)
	OLS	IV	IV	IV
	$\ln(1 + C_{o,d}^{1980,2019})$			
$\ln(1 + A_{o,d,1980}^{ng})$	0.525*** (0.041)	0.820*** (0.116)	0.833*** (0.116)	1.064*** (0.106)
Observations	62,820	62,820	62,820	62,820
Origin fixed effects	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes
Distance polynomials	No	No	Yes	Yes
Latitude polynomials	No	No	Yes	Yes
Origin $\times$ region fixed effects	No	No	No	Yes
KP first-stage F		87.4	58.6	70.8
Stock-Yogo (10% size)		31.50	31.50	31.50
Stock-Yogo (5% rel. bias)		19.86	19.86	19.86
Anderson–Rubin F		35.0	35.6	19.6
Anderson–Rubin p		0.000	0.000	0.000

Notes: This table repeats the estimation of Table 4 on the 20 origin countries that ever record a positive child outcome. Column 1 reports OLS estimates and columns 2–4 instrument absorption using the seven push–pull census vintages. Column 3 adds cubic controls for origin–destination distance and latitude difference, and column 4 adds origin $\times$ destination–region fixed effects. All specifications include origin and destination fixed effects. Standard errors (in parentheses) are clustered by destination state. * $p < .10$, ** $p < .05$, *** $p < .01$.

C.6 Stability over Time

Figure C.1: The causal effect of foreign ideas on U.S. innovation, before and after 2000.



Notes: The figure plots coefficient estimates with 95% confidence intervals from two regressions of the Figure 4 design with the invention years split at 2000. The first regresses $\ln(1 + C_{o,d}^{1980,2000})$ on the 1980 stock $\ln(1 + A_{o,d,1980}^{ng})$ and the second $\ln(1 + C_{o,d}^{2000,2019})$ on the 2000 stock $\ln(1 + A_{o,d,2000}^{ng})$, each in the specification of column 3 of Table 4. The reported test of the equality of the two coefficients comes from a stacked system in which the regressor, the excluded instruments, the geography polynomials, and the fixed effects are interacted with the period. The confidence intervals use standard errors clustered by destination state.

C.7 Inference

Table C.11: The causal effect of foreign ideas on U.S. innovation, alternative clustering.

	(1)	(2)	(3)	(4)
	OLS	OLS	IV	IV
<i>Panel A. Flow of US inventions with parents from origin o</i>				
	$\ln(1 + C_{o,d}^{1980,2019})$		$\ln(1 + C_{o,d}^{P,1980,2019})$	
$\ln(1 + A_{o,d,1980}^{ng})$	0.749** (0.040)	0.749** (0.040)	1.432** (0.103)	2.566** (0.169)
<i>Panel B. Flow of US inventions with parents from origin o</i>				
	$\ln(1 + C_{o,d}^{1980,2019})$		$\ln(1 + C_{o,d}^{P,1980,2019})$	
$\ln(1 + A_{o,d,1980}^{P,ng})$	0.533** (0.026)	0.533** (0.026)	0.944** (0.060)	1.690** (0.099)
Observations	606,213	606,213	606,213	606,213
Origin fixed effects	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes
Distance polynomials	No	Yes	Yes	Yes
Latitude polynomials	No	Yes	Yes	Yes
Standard errors clustered by	County	County	County	County

Notes: This table repeats Table 4, changing only the level at which standard errors are clustered, from destination state, 51 clusters, to county, 3,141 clusters. Variables, specification, sample and instruments are those of Table 4. * $p < .10$, ** $p < .05$, *** $p < .01$.

D Appendix to Section 5

D.1 Timing of Inputs Availability

Table D.1: The causal effect of foreign inputs on U.S. innovation, timing of the estimation sample.

	(1)	(2)	(3)	(4)
	IV	IV	IV	IV
	<i>US invention with foreign inputs</i>			
	$I_{d,t}(\tau')$			
All foreign inputs available at $t - 1$ (modal country)	0.227*** (0.026)	0.214*** (0.038)	0.132*** (0.025)	0.105*** (0.036)
Observations	19,214,020	10,494,257	10,470,772	2,412,384
Origin fixed effects	Yes	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes	Yes
Periods after the county invents	Yes	Yes	No	No
Periods after the founding window	Yes	No	No	No
Invention rate in the sample	0.0080	0.0052	0.0030	0.0069
KP first-stage F	95.4	33.3	27.0	31.0

Notes: This table repeats column 4 of Table 7, changing only the estimation sample. Column 1 is that column. Column 2 keeps only the periods in which the technology can still be invented, those overlapping the interval from its first patent to five years after its birth. Column 3 additionally drops, for each technology and county, the periods after the county has invented, so that the outcome is the chance of inventing among those that have not yet done so. Column 4 keeps, for each technology, only the five-year period containing its birth year, one observation per county, so that the period fixed effect becomes a birth-cohort effect. The invention rate row gives the mean of the dependent variable in each column's own sample. Variables, specifications and instruments are those of Table 7. Standard errors (in parentheses) are clustered by technology. * $p < .10$, ** $p < .05$, *** $p < .01$.

D.2 Inference

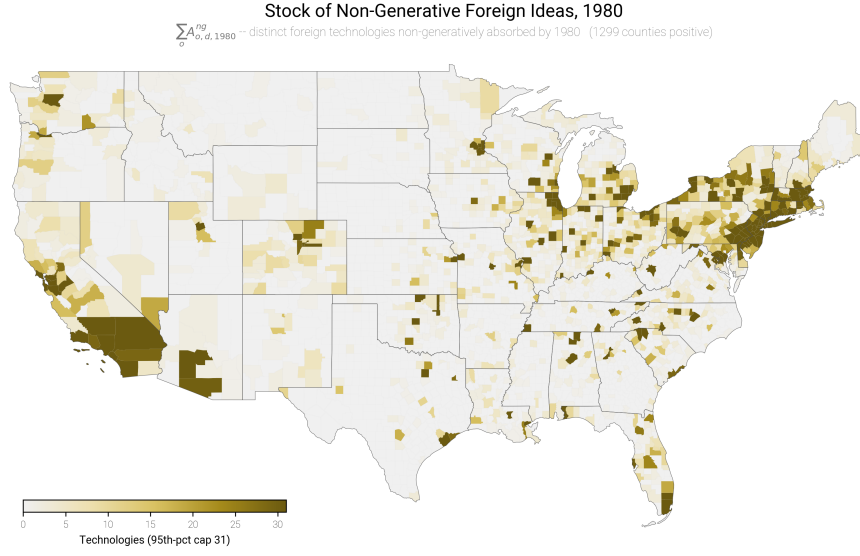
Table D.2: The causal effect of foreign inputs on U.S. innovation, alternative clustering.

	(1)	(2)	(3)
	IV	IV	IV
	<i>US invention with foreign inputs</i>		
	$I_{d,t}(\tau')$		
All foreign inputs available at $t - 1$ (modal country)	0.227*** (0.026)	0.227*** (0.054)	0.227*** (0.073)
Observations	19,214,020	19,214,020	19,214,020
Origin fixed effects	Yes	Yes	Yes
Destination fixed effects	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes
Distance polynomials	Yes	Yes	Yes
Standard errors clustered by	Technology	County	State

Notes: This table repeats column 4 of Table 7, changing only the level at which standard errors are clustered. Column 1 is that column, clustered by technology, 1,250 clusters. Column 2 clusters by county, 1,933 clusters, and column 3 by destination state, 51 clusters. The estimate is identical across the three columns; only its standard error differs. Variables, specification and instruments are those of Table 7. * $p < .10$, ** $p < .05$, *** $p < .01$.

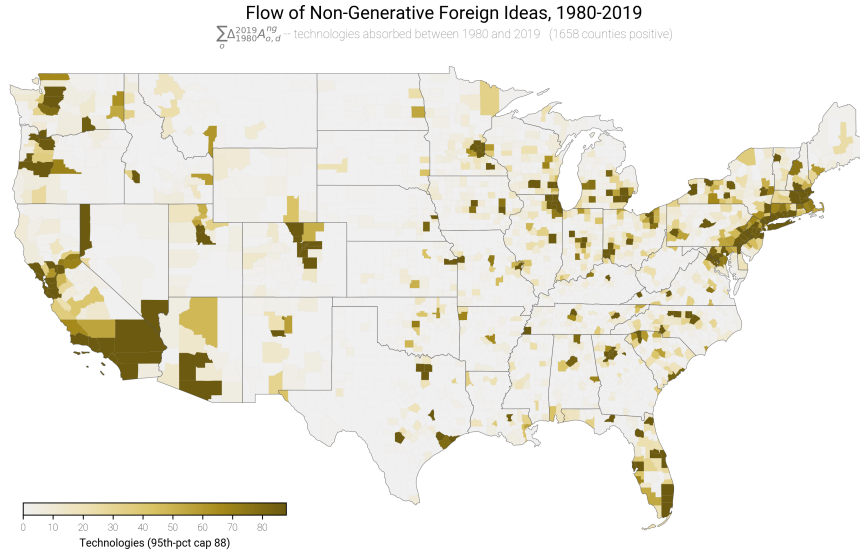
D.3 The Geography of the Stock of Foreign Ideas

Figure D.1: Stock of non-generative foreign ideas, 1980.



Notes: Each county's stock of non-generatively absorbed foreign technologies in 1980, $\sum_o A_{o,d,1980}^{ng}$, the regressor of Table 4 summed over origins.

Figure D.2: Flow of non-generative foreign ideas, 1980–2019.



Notes: Each county's flow of non-generatively absorbed foreign technologies between 1980 and 2019, $\sum_o \Delta_{1980}^{2019} A_{o,d}^{ng}$, the outcome of Panel C column 4 of Table 3 summed over origins.